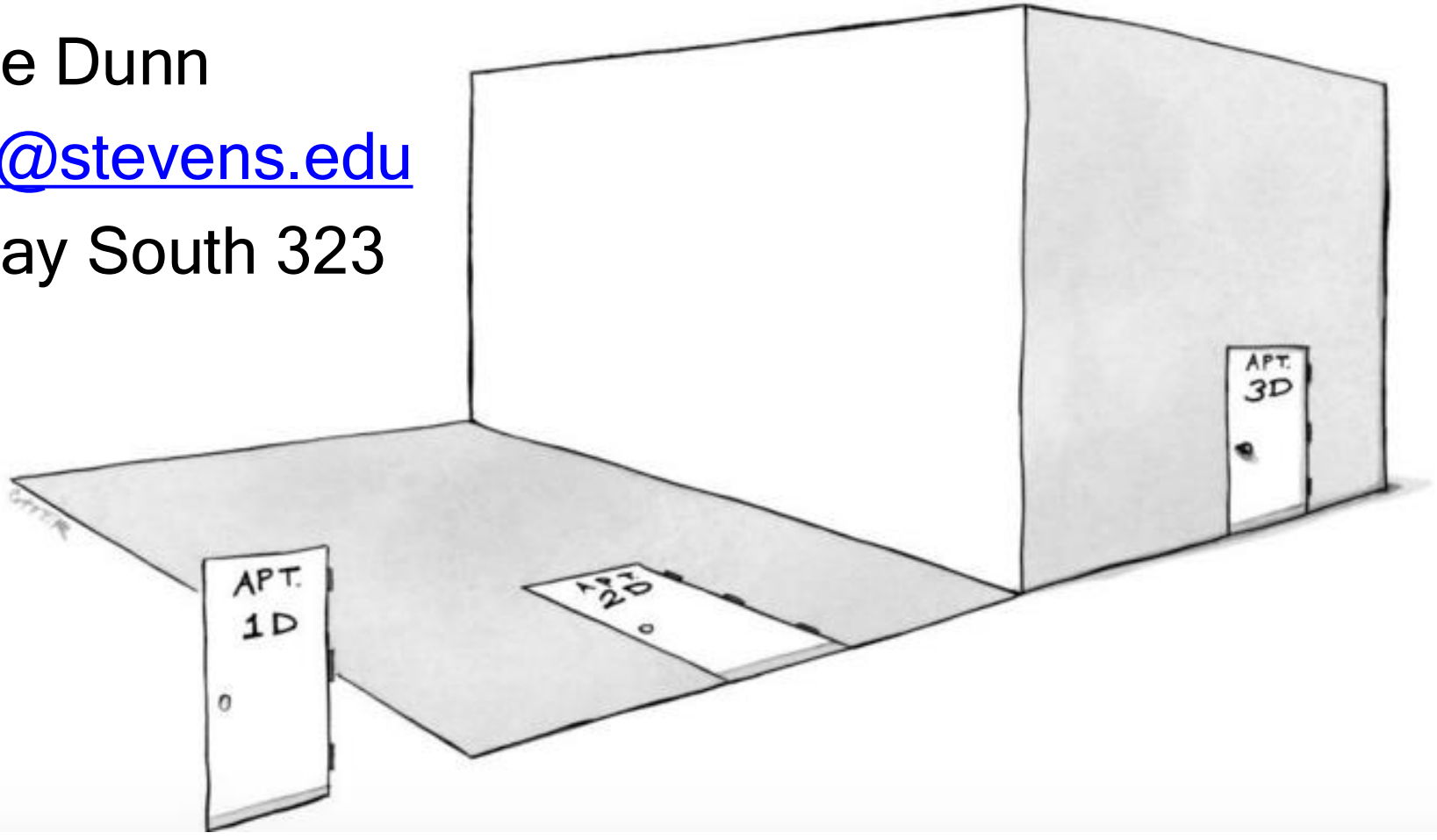
CS 532: 3D Computer Vision
Lecture 1

Enrique Dunn

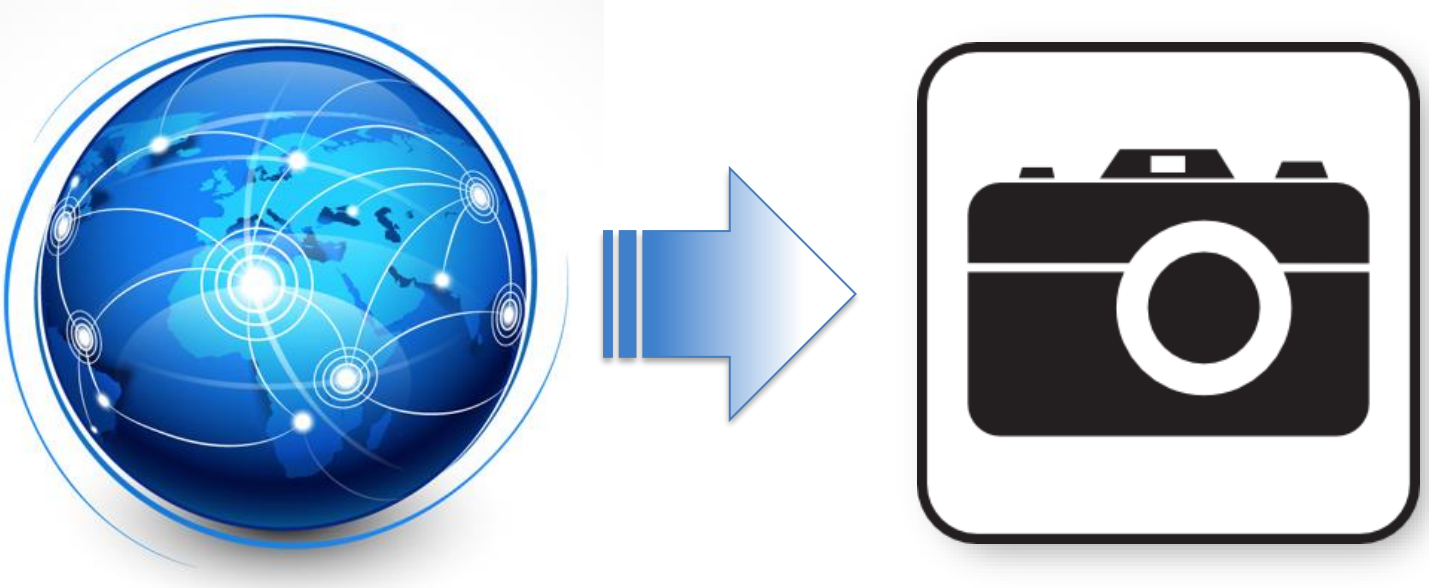
edunn@stevens.edu

Gateway South 323



What if …

we could turn the Internet into a camera?



Uploads
per minute



facebook 130,000 Images

YouTube 300 hrs. of video

Visual Index of the World



Visual Index of the World

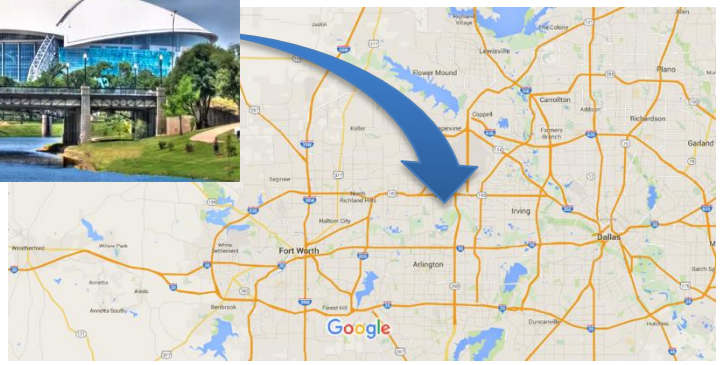


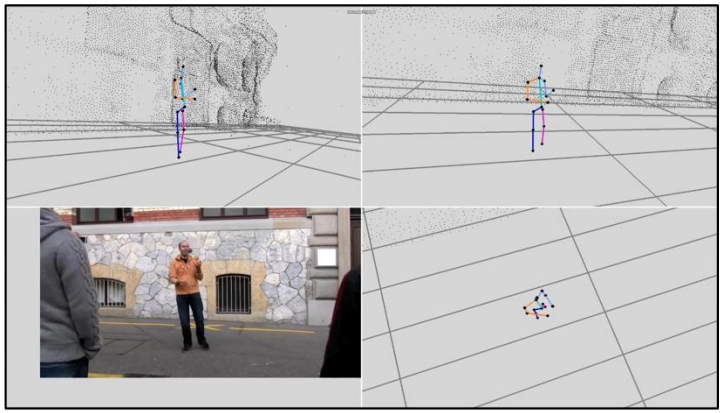
Applications











Objectives

- Approach Computer Vision from a geometric, 3D perspective
  - Negligible overlap with traditional Computer Vision course (CS 558)
  - Explain image formation, single and multi-view geometry, structure from motion
  - Reconstruct and represent dense 3D geometry
- Introduce Computational Geometry concepts
  - Point clouds, meshes, Delaunay triangulation

Important Points

- This is an elective course. You chose to be here.
- If you are in class… be in class
- Expect to work and to be challenged.
- Exams won't be based on recall. They will be open-book and you will be expected to solve new problems.

Logistics

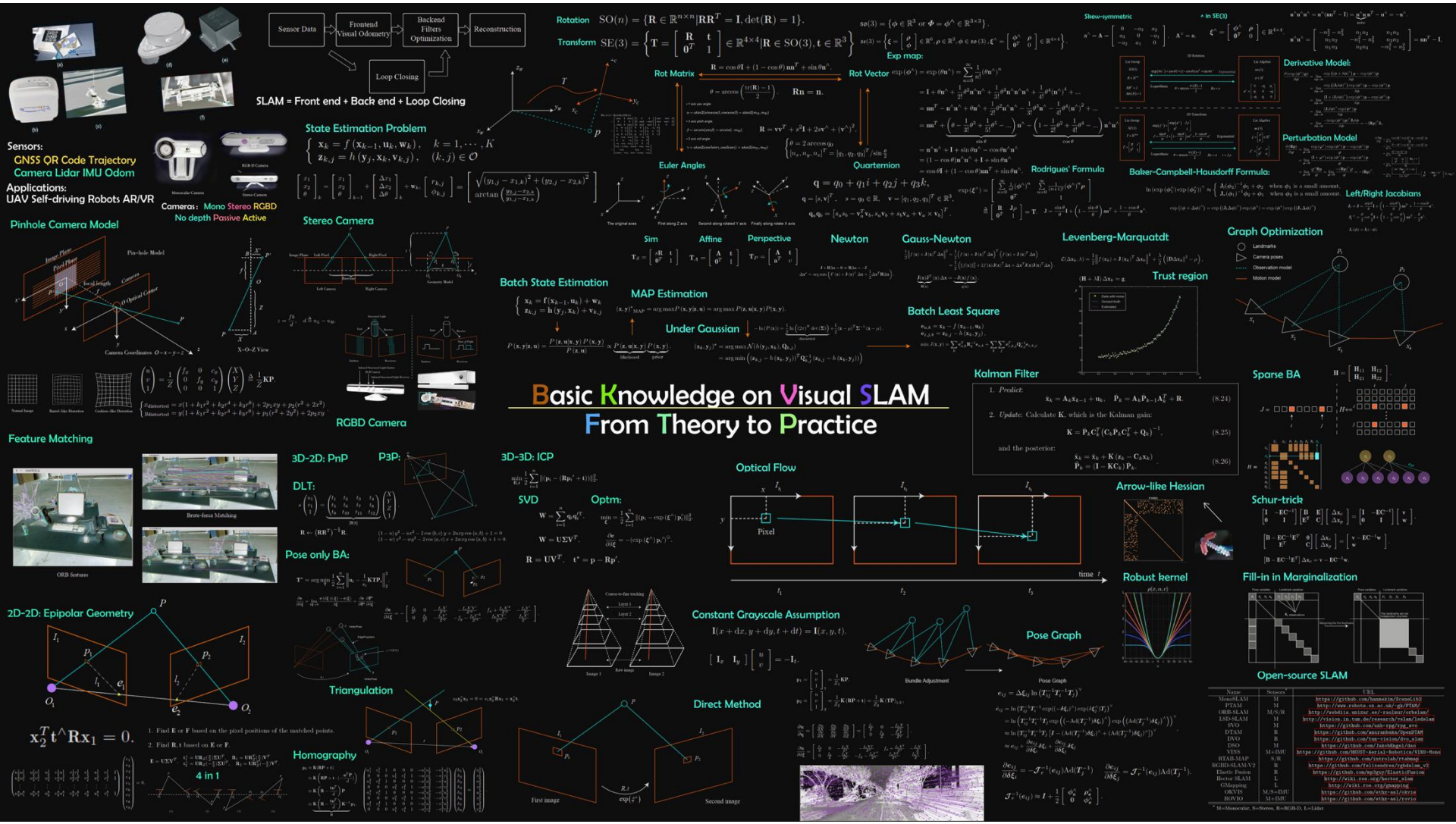
- Office hours:

  Wednesday 5-6 @ GS323
- Evaluation:
  - 4 homework sets (50%)
  - Mid-term exam (20%) March 3rd
  - Final exam (25%)
  - Weekly quizzes (5%)

Textbooks

- Richard Szeliski, Computer Vision: Algorithms and Applications, Springer, 2010
- David M. Mount, CMSC 754: Computational Geometry lecture notes, Department of Computer Science, University of Maryland, Spring 2012
- Xiang Gao and Tao Zhang Introduction to Visual SLAM From Theory to PracticeLinks to an external site.
- All available online

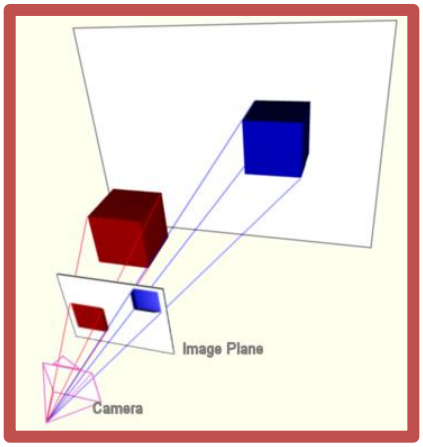
Introduction to Visual SLAM From Theory to PracticeLinks to an external site.

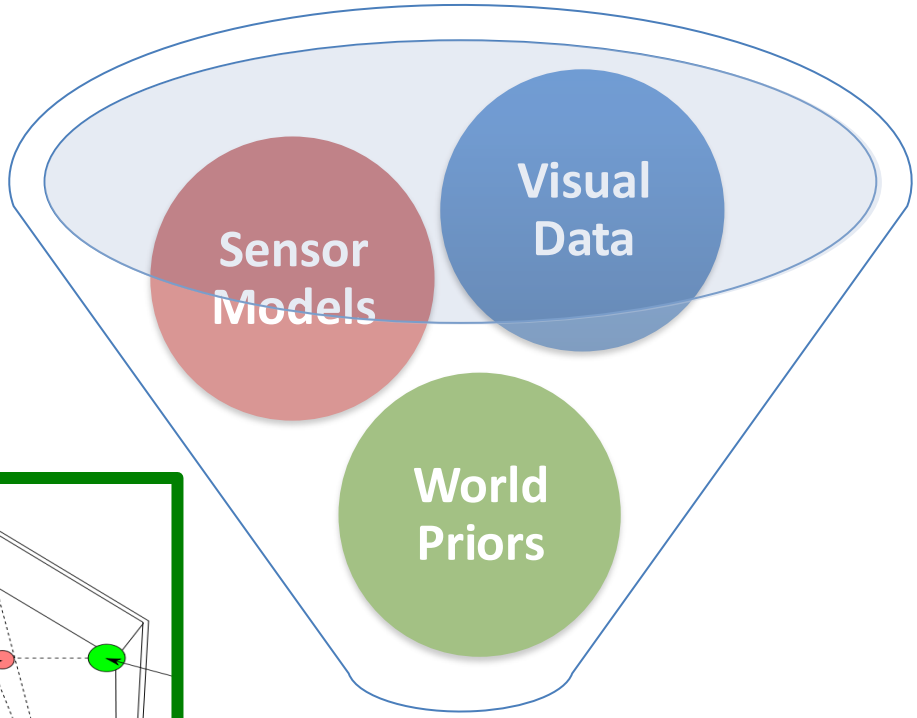


What is Computer Vision

- Extracting Concepts from Pixels

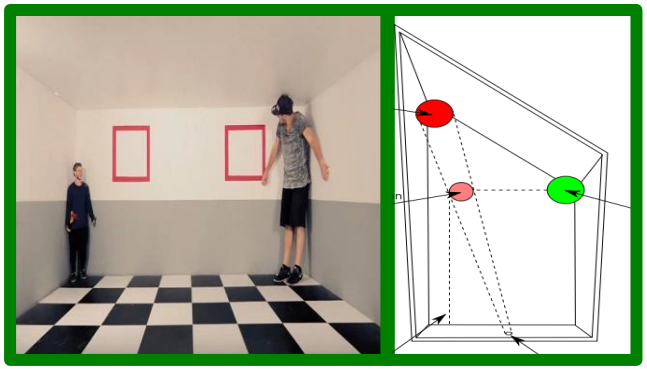
Computer Vision





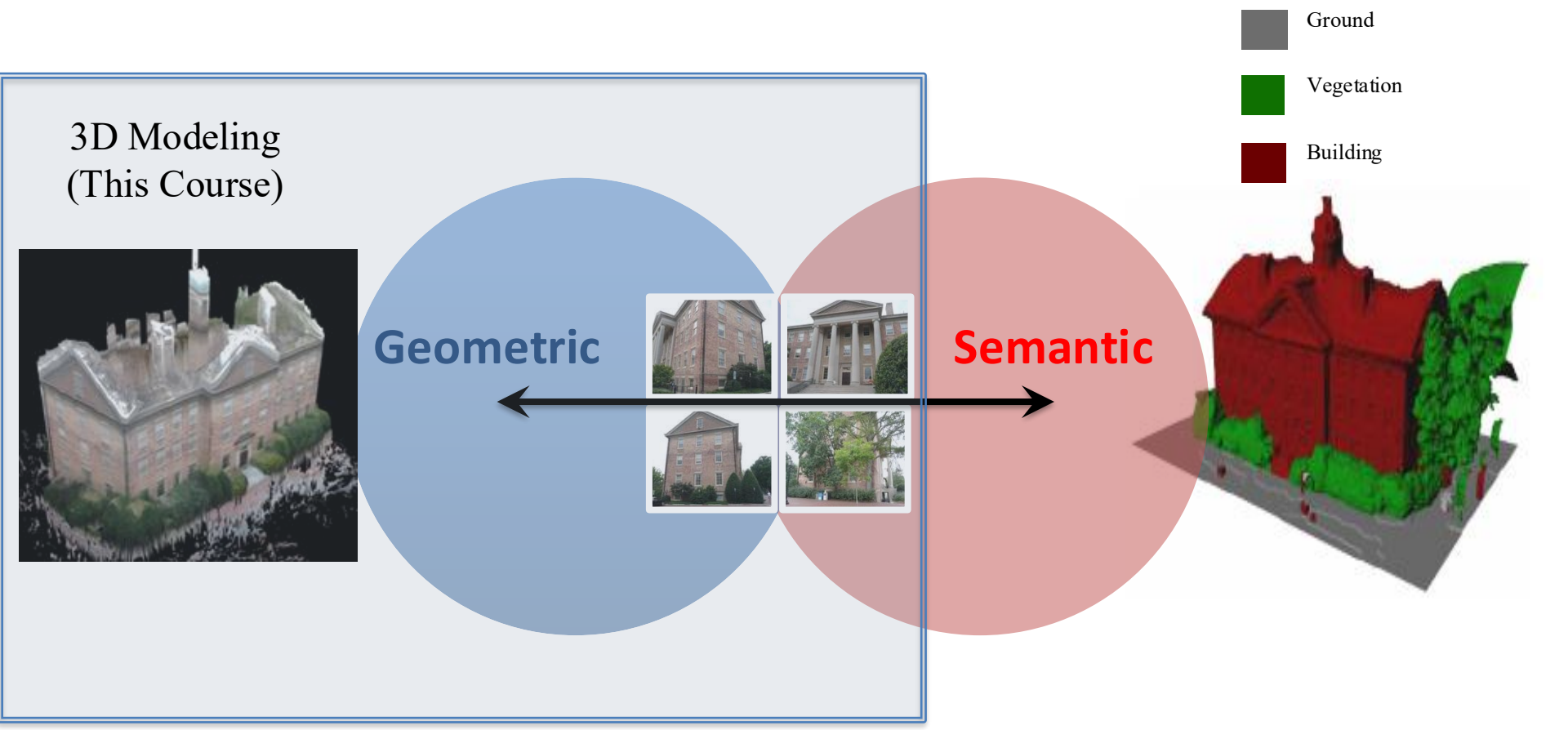






Visual Concepts

Visual Concepts



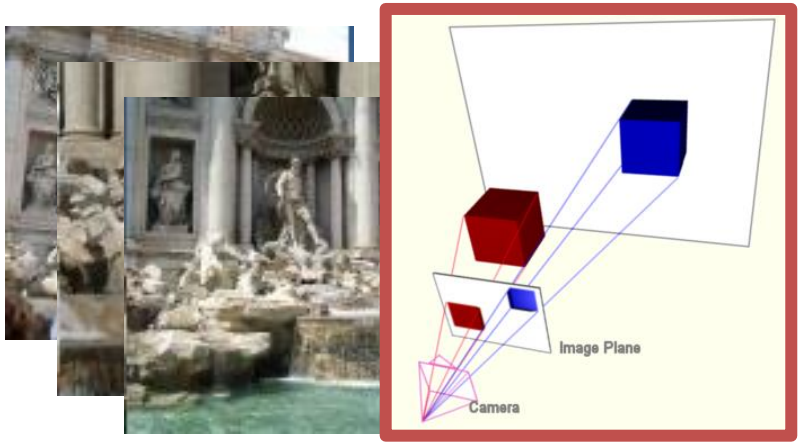
3D Modeling Problem Statement





Recover the 3D geometry and appearance of an environment from 2D images/videos captured from multiple viewpoints.

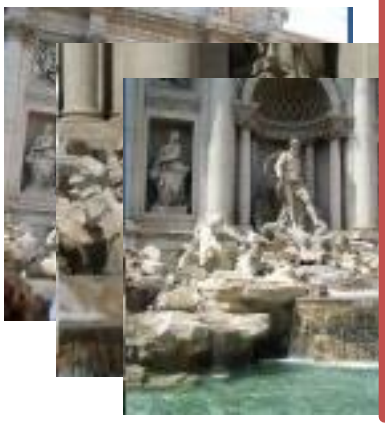
3D Modeling Problem Statement

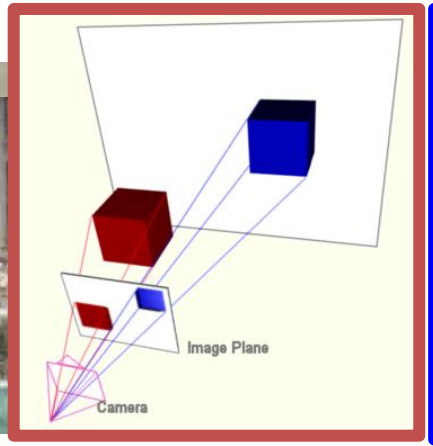


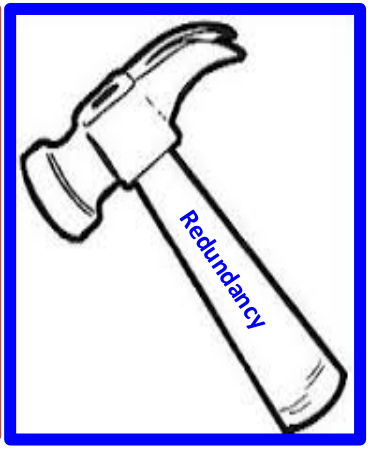


Recover the 3D geometry and appearance of an environment from 2D images/videos captured from multiple viewpoints.

3D Modeling Problem Statement



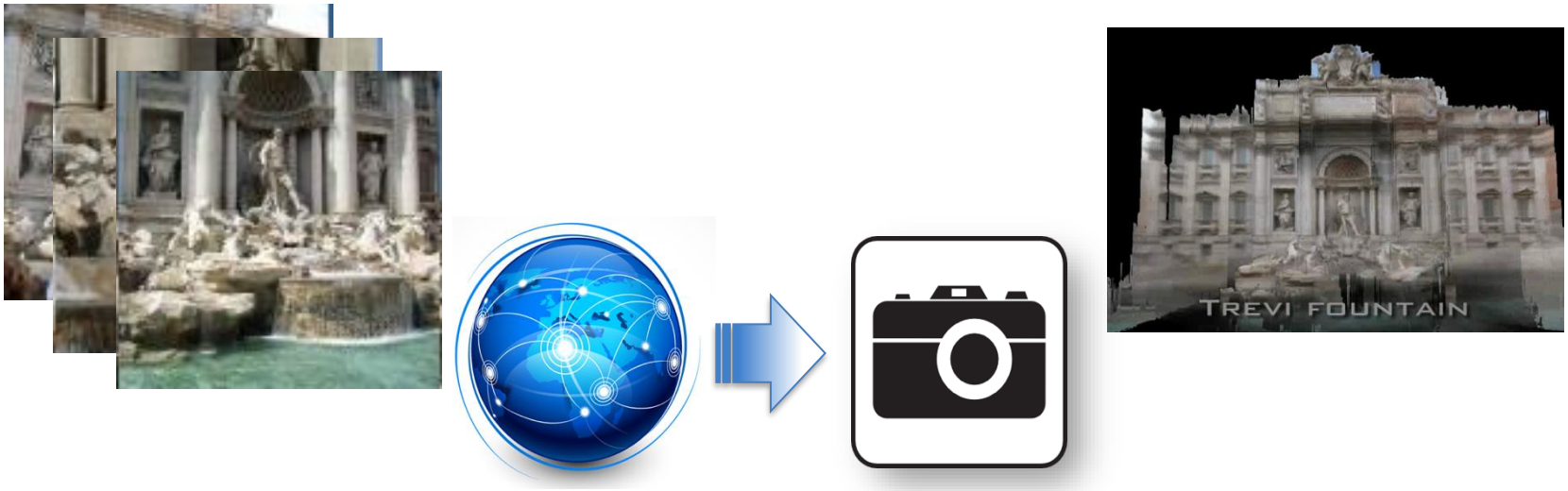






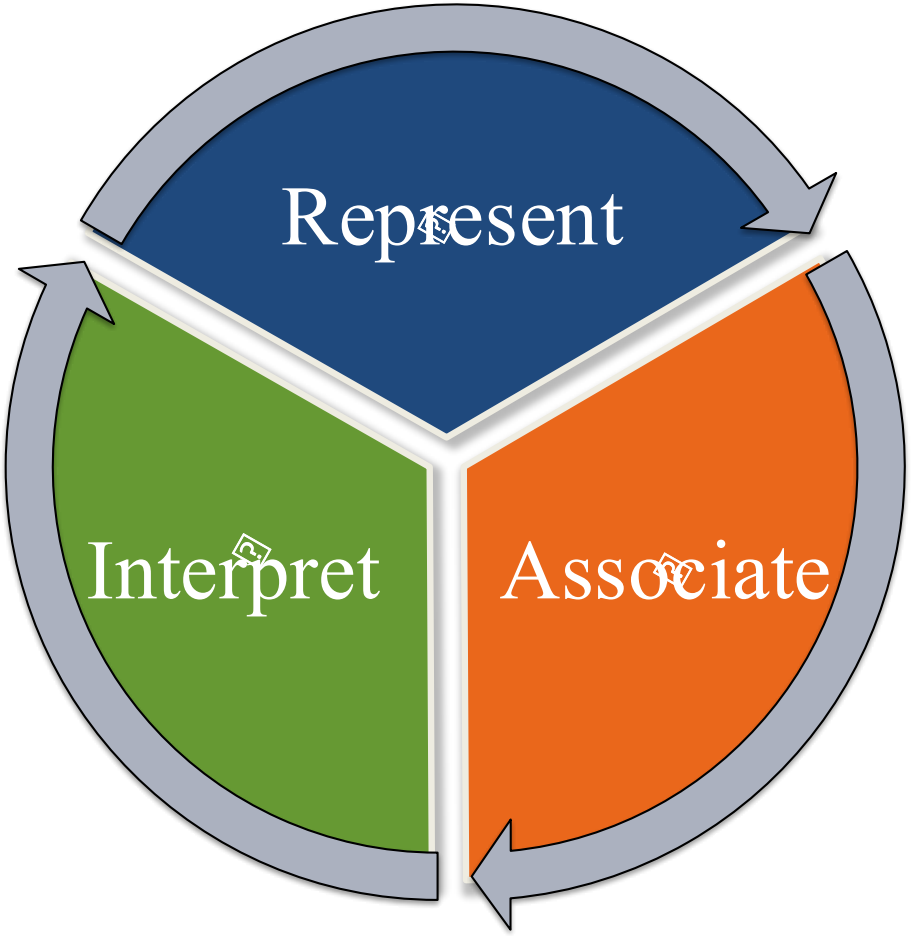
Recover the 3D geometry and appearance of an environment from 2D images/videos captured from multiple viewpoints.

Data-Driven 3D Modeling



Data Inclusive
Content Extensive
High Fidelity

The 3D Modeling Process



SfM: Data Association

Spatial ordering

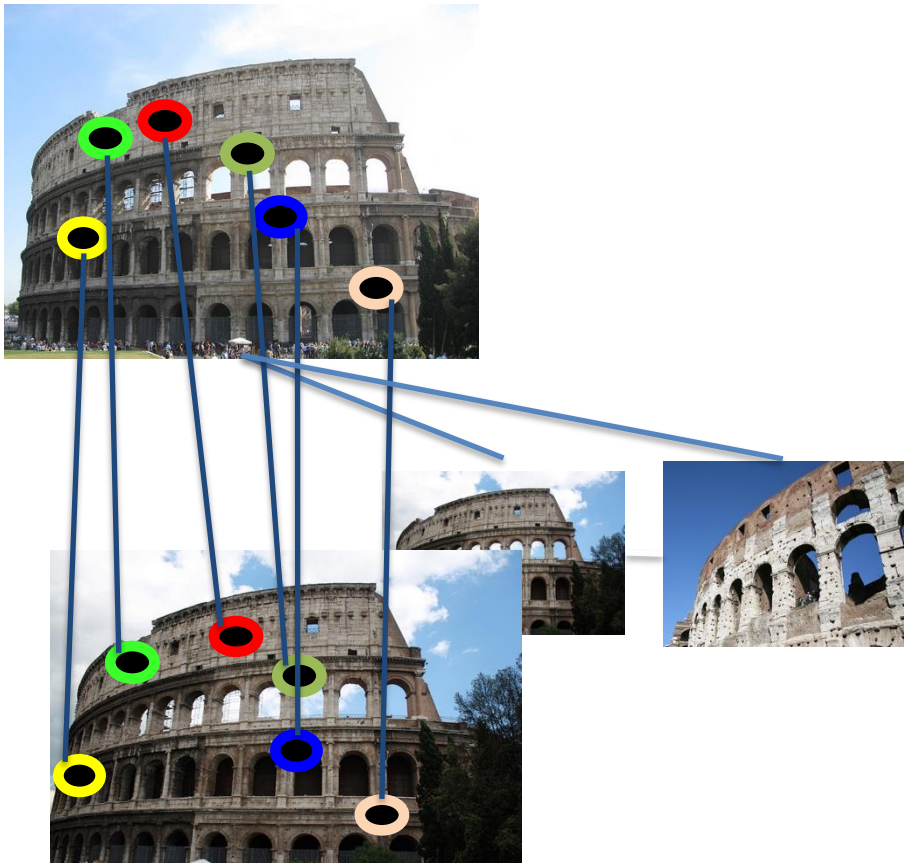
images, 2D features



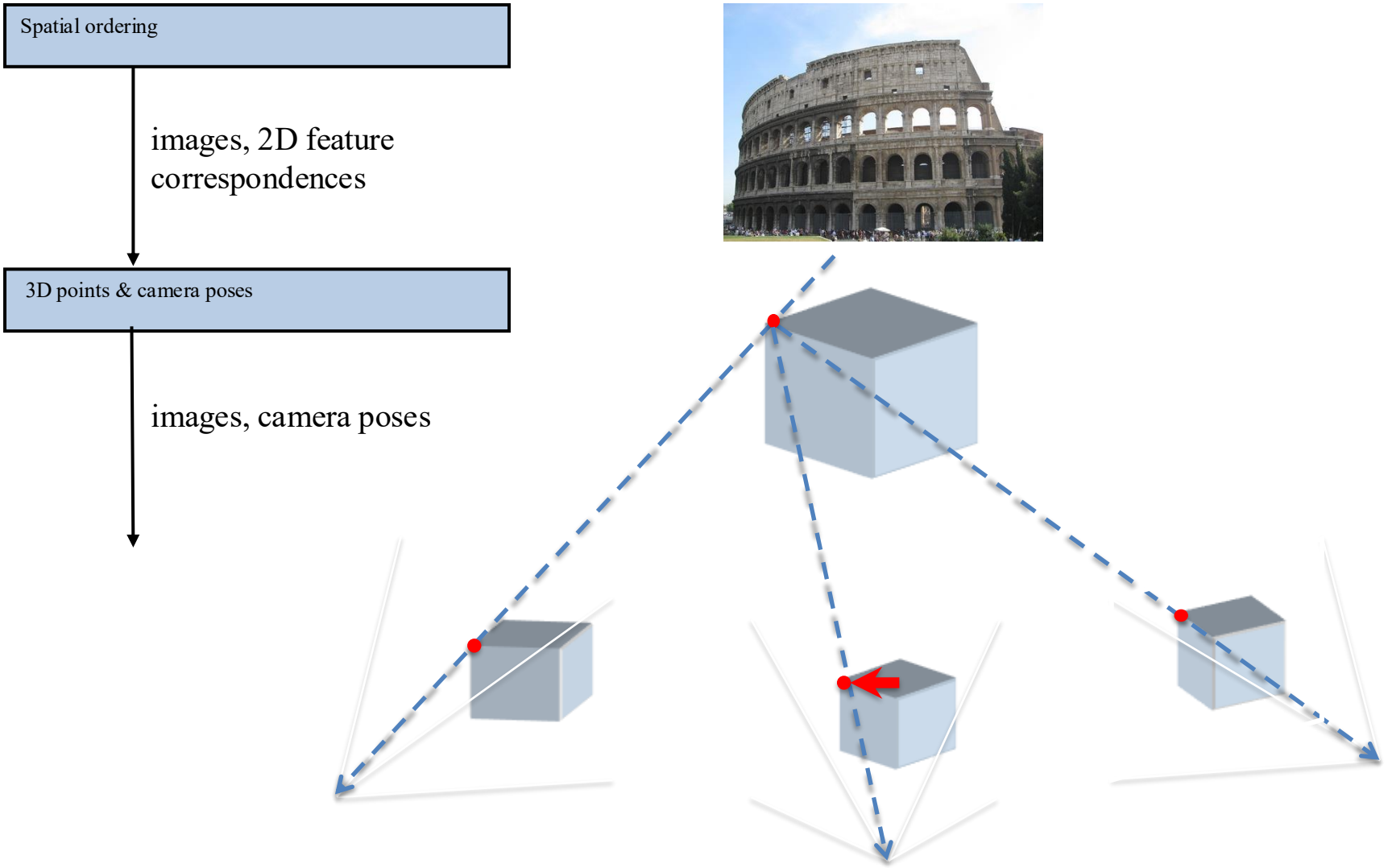
SfM: Registration

Spatial ordering

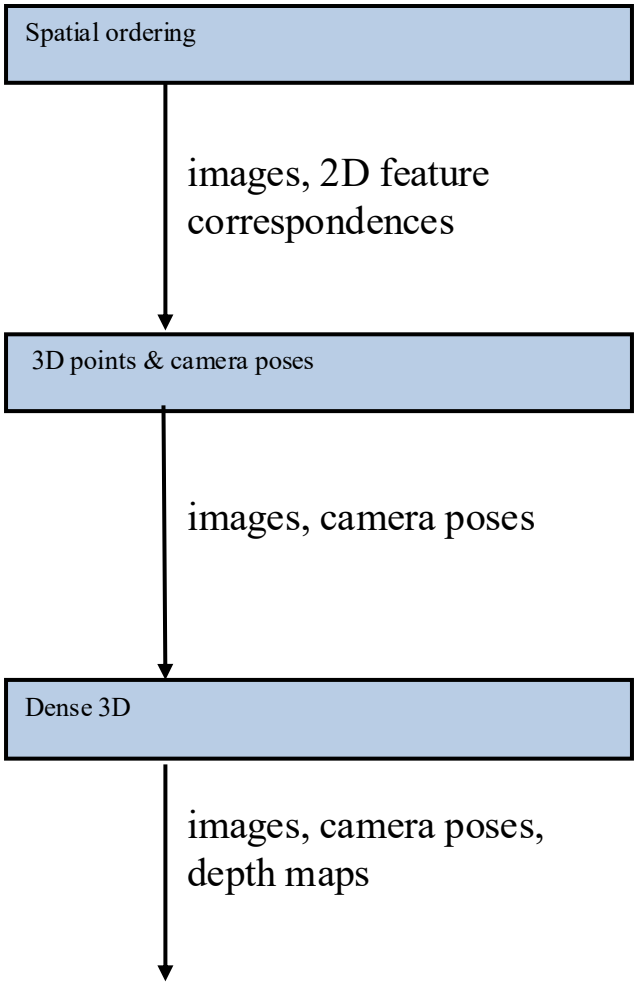
images, 2D features
correspondences



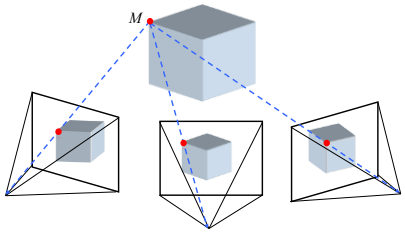
SfM: Registration

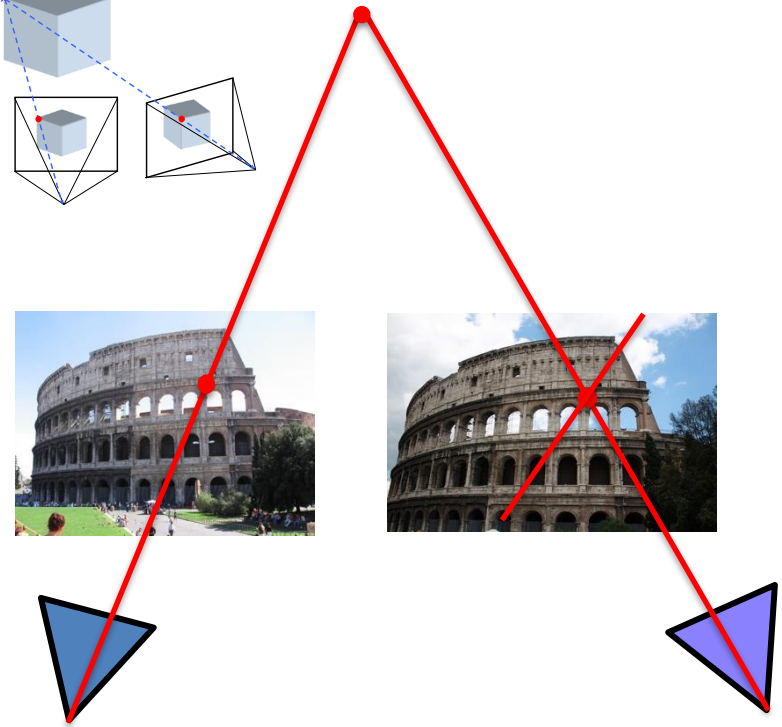


Dense 3D Modeling

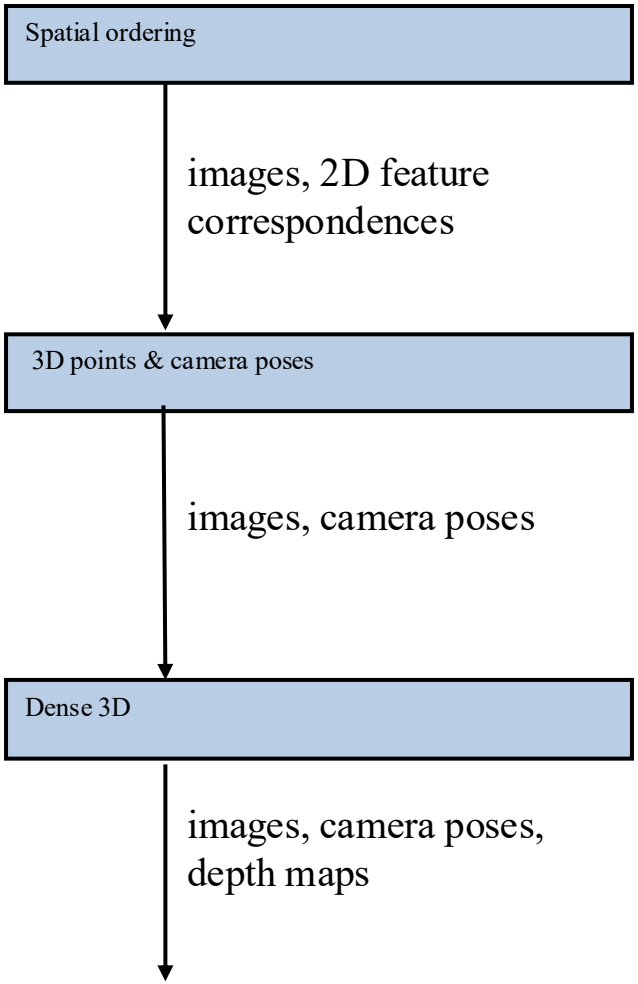




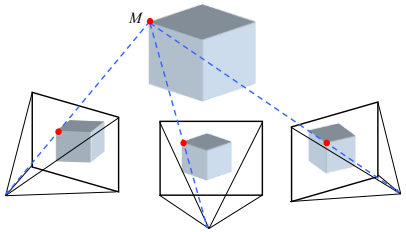


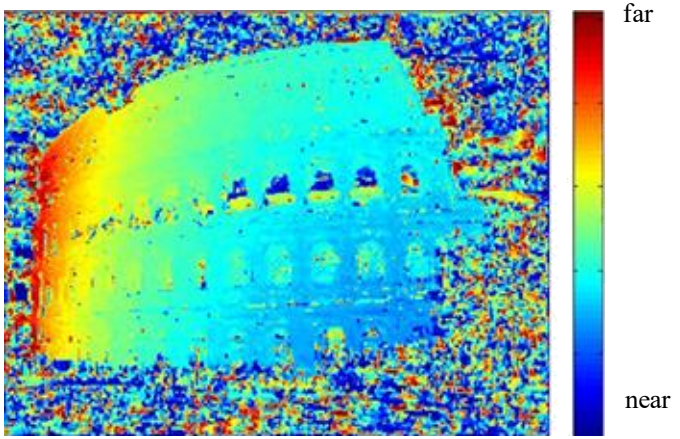


3D model from Images

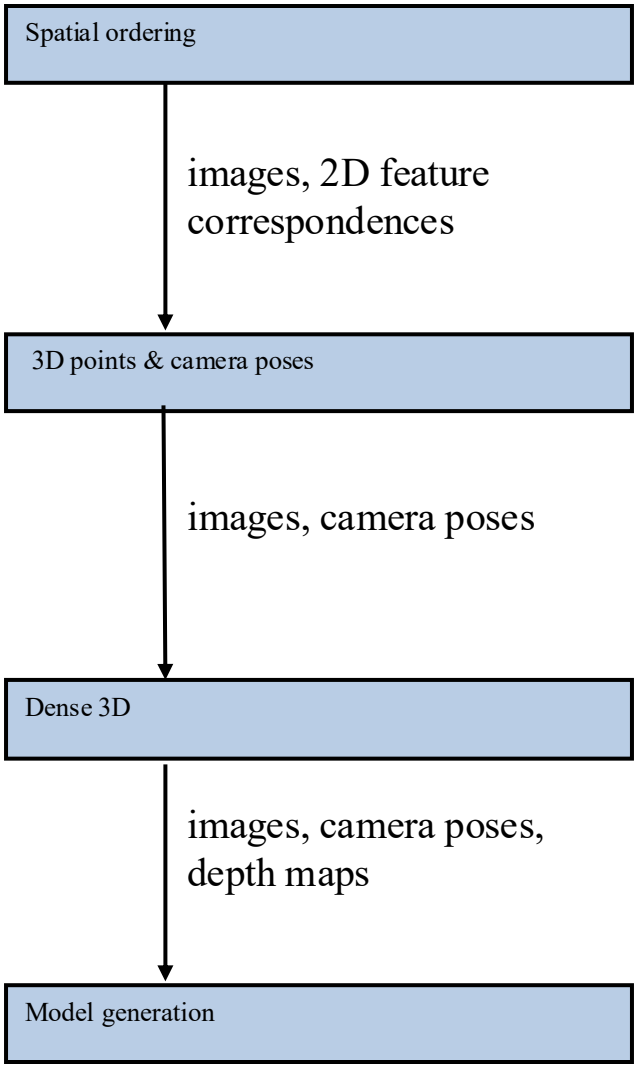




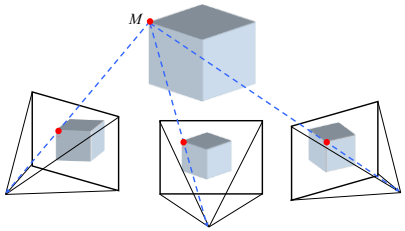


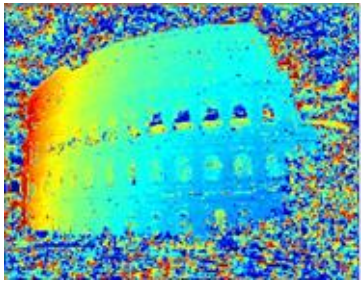


3D Model Generation



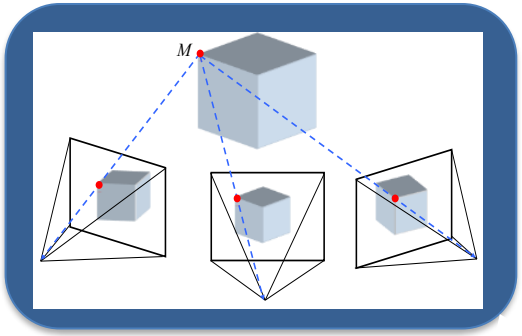




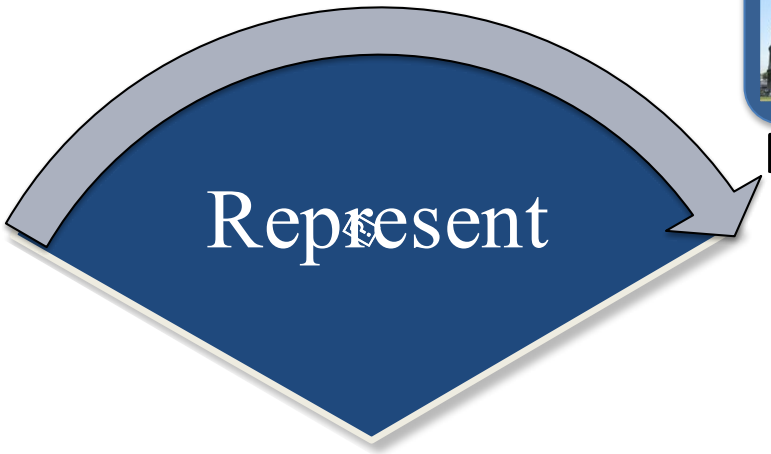


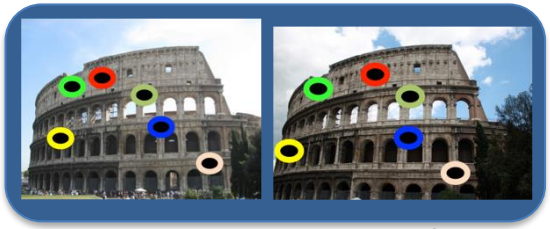


The 3D Modeling Process

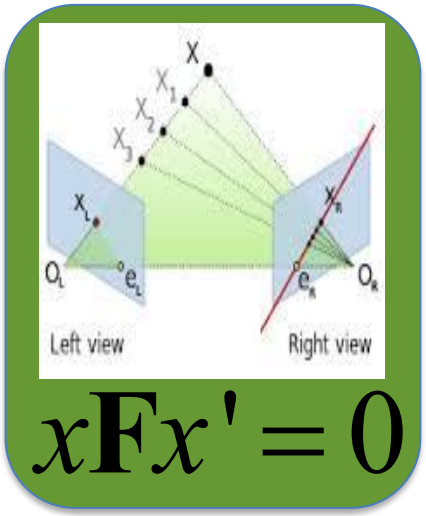


Camera Pose

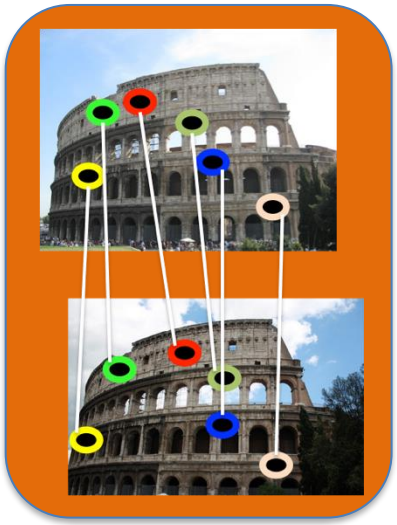




Feature Extraction

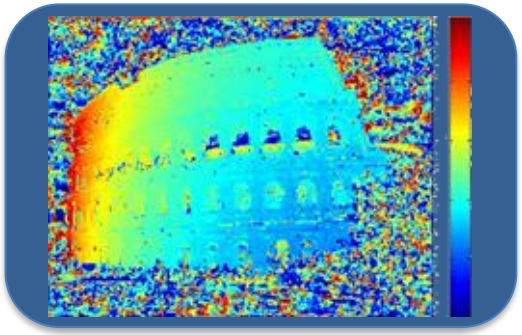


Epipolar Geometry



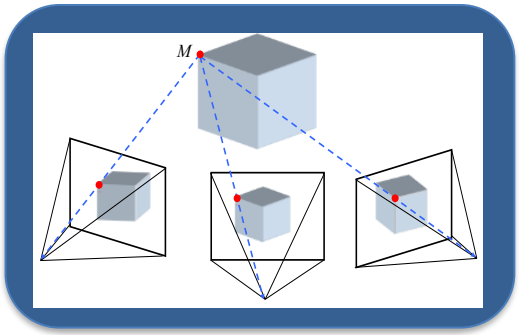
Feature Matching

The 3D Modeling Process

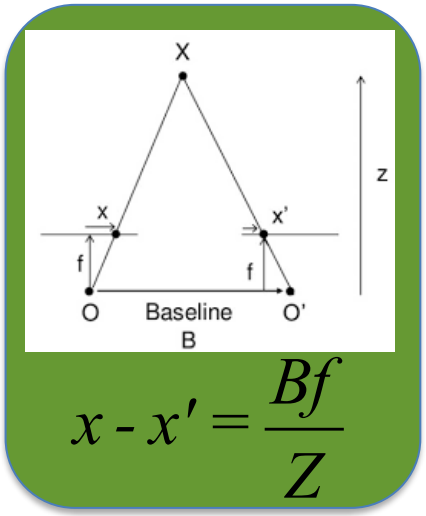


Depth-Map

Represent

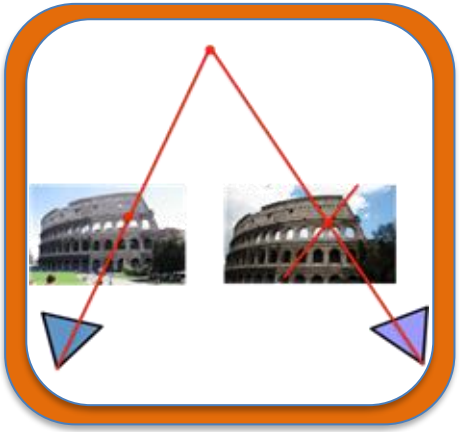


Camera Pose



$$x - x' = \frac{Bf}{Z}$$

Depth from Disparity

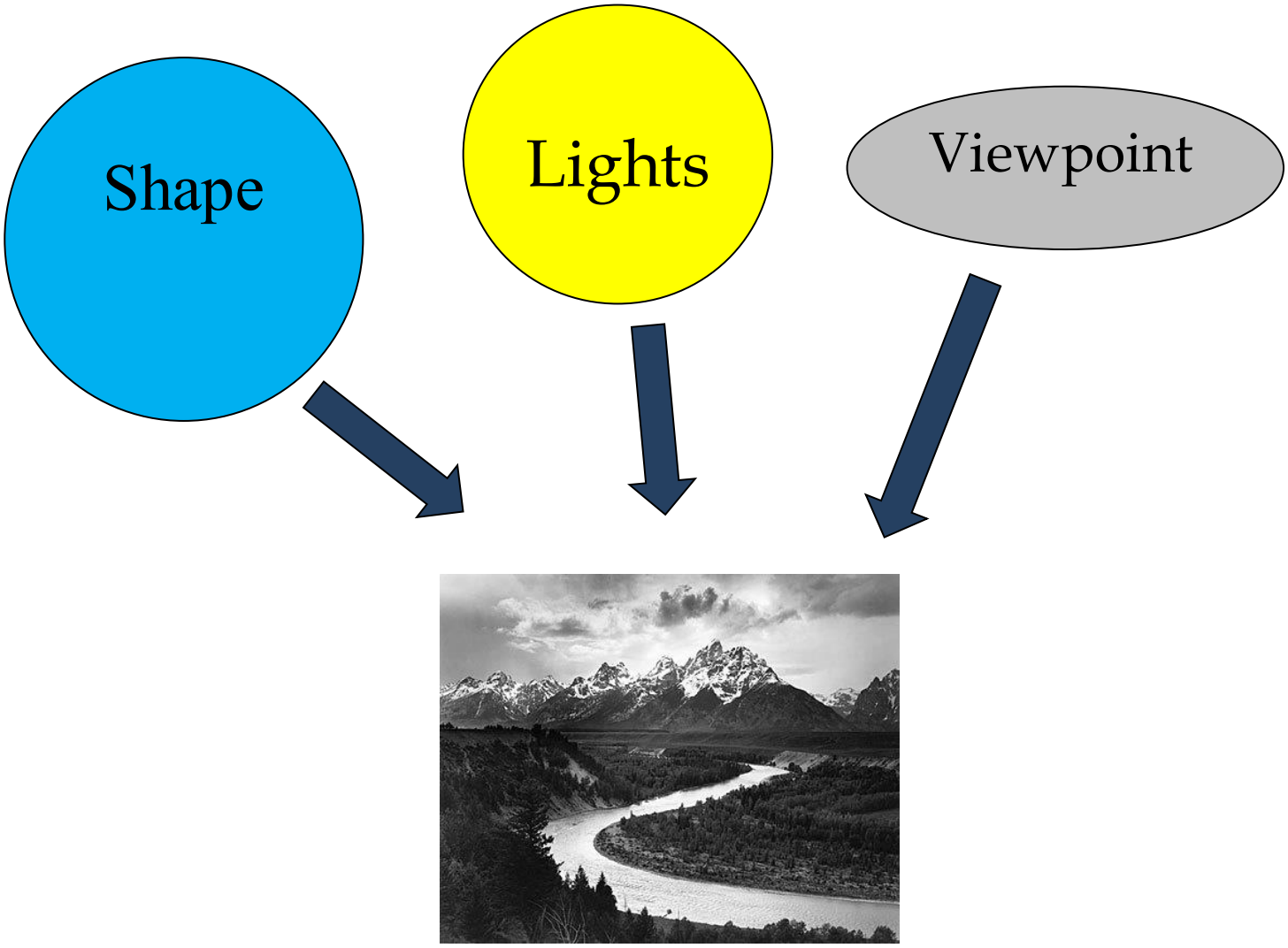


Pixel Correspondences

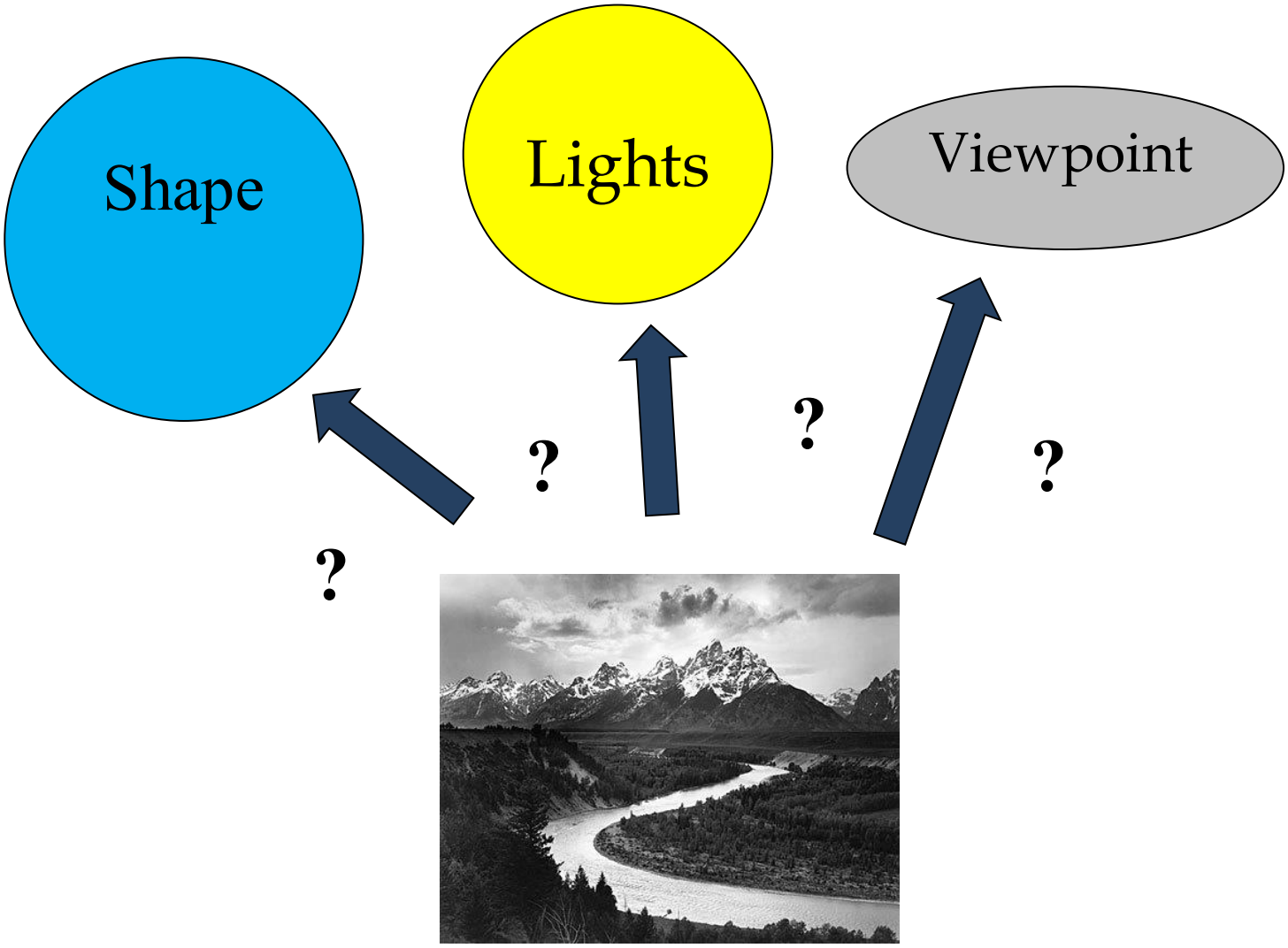
What is Computer Vision

- It is not image processing
- It is not graphics

Graphics vs. Vision



Graphics vs. Vision



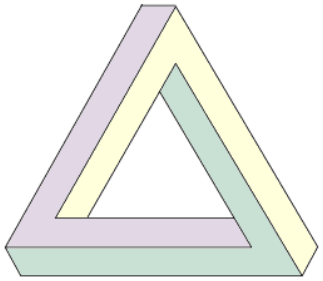
Vision is Hard

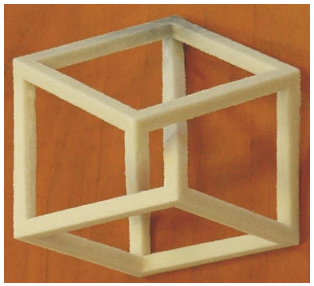


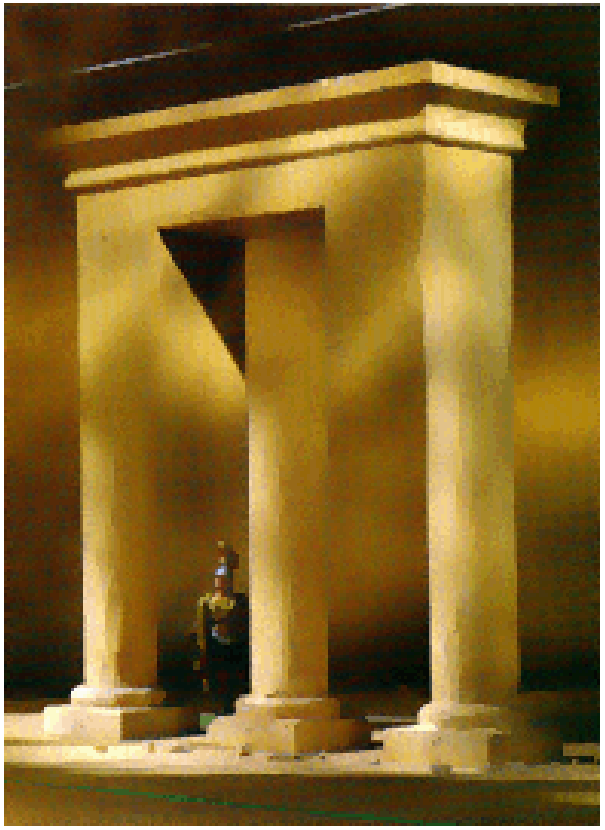
Vision is Hard

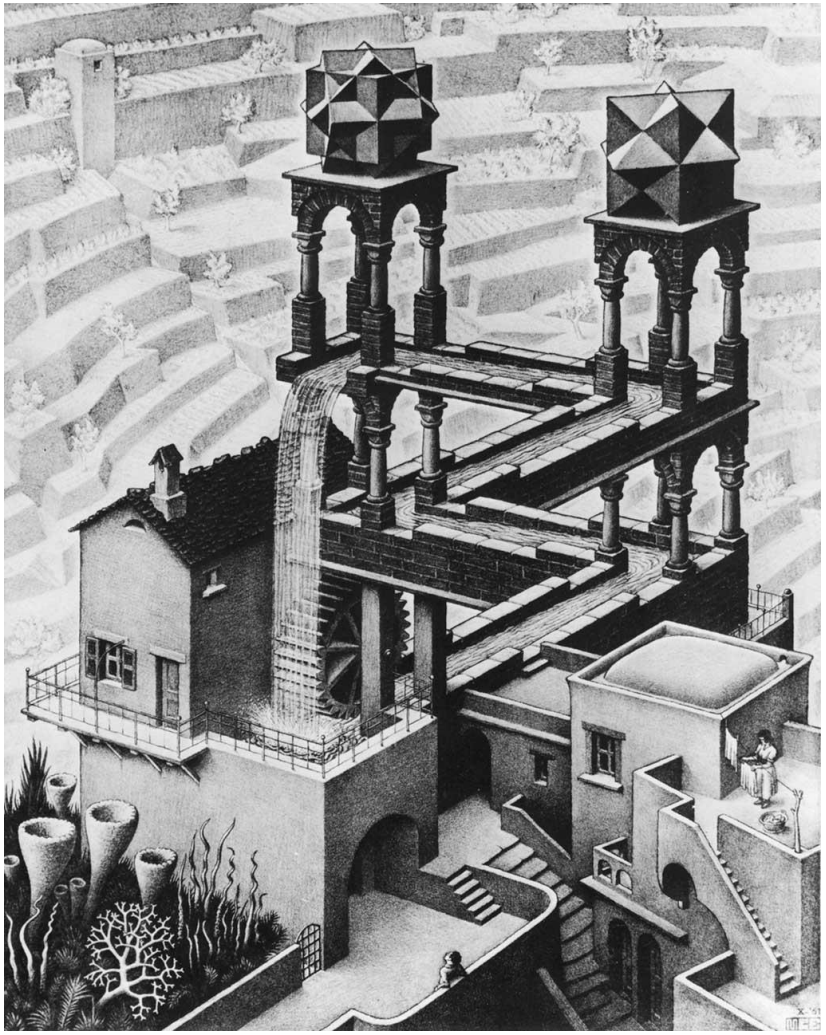


Vision is Hard

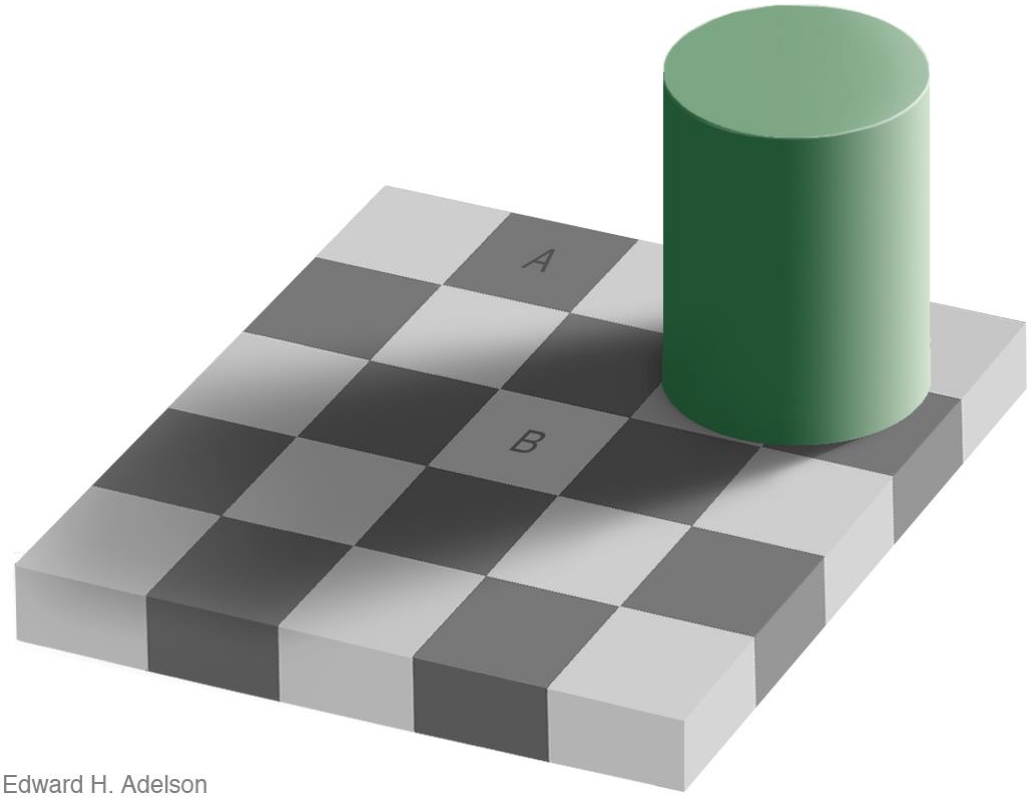






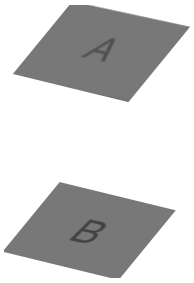


Vision is Hard

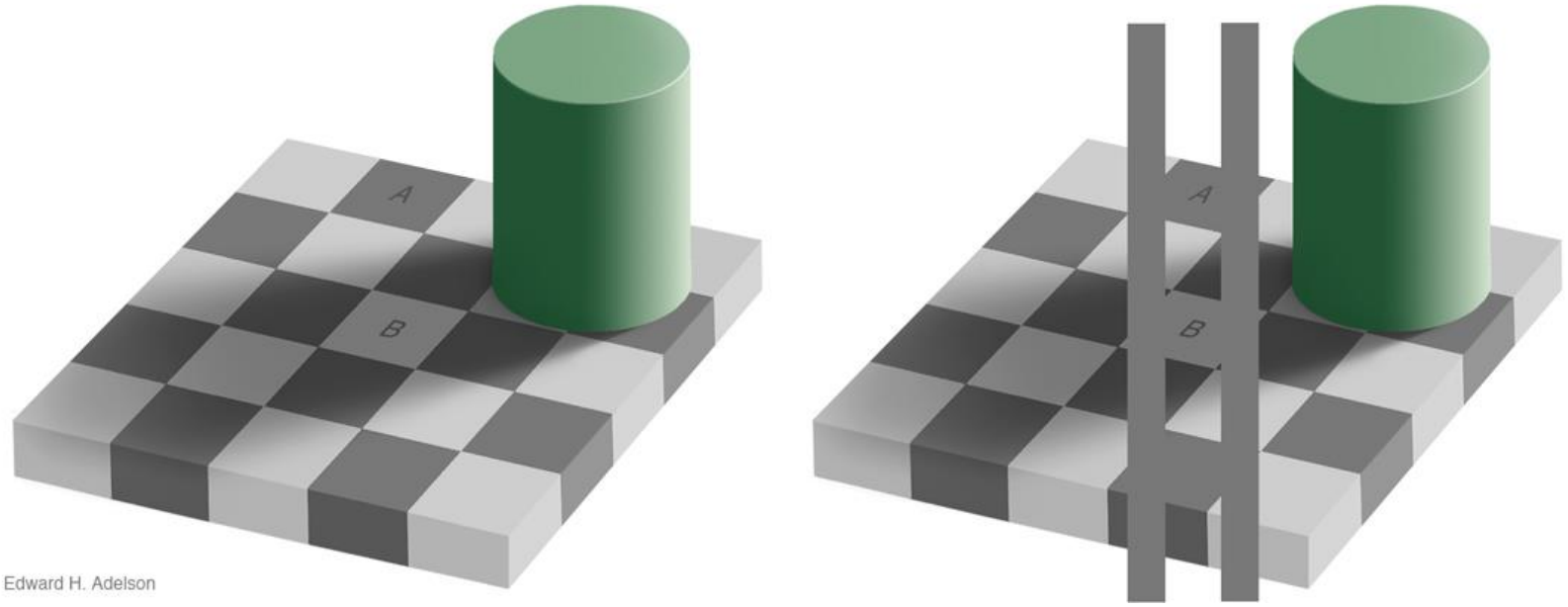


Edward H. Adelson

Vision is Hard



Vision is Hard



Edward H. Adelson

Vision is Hard



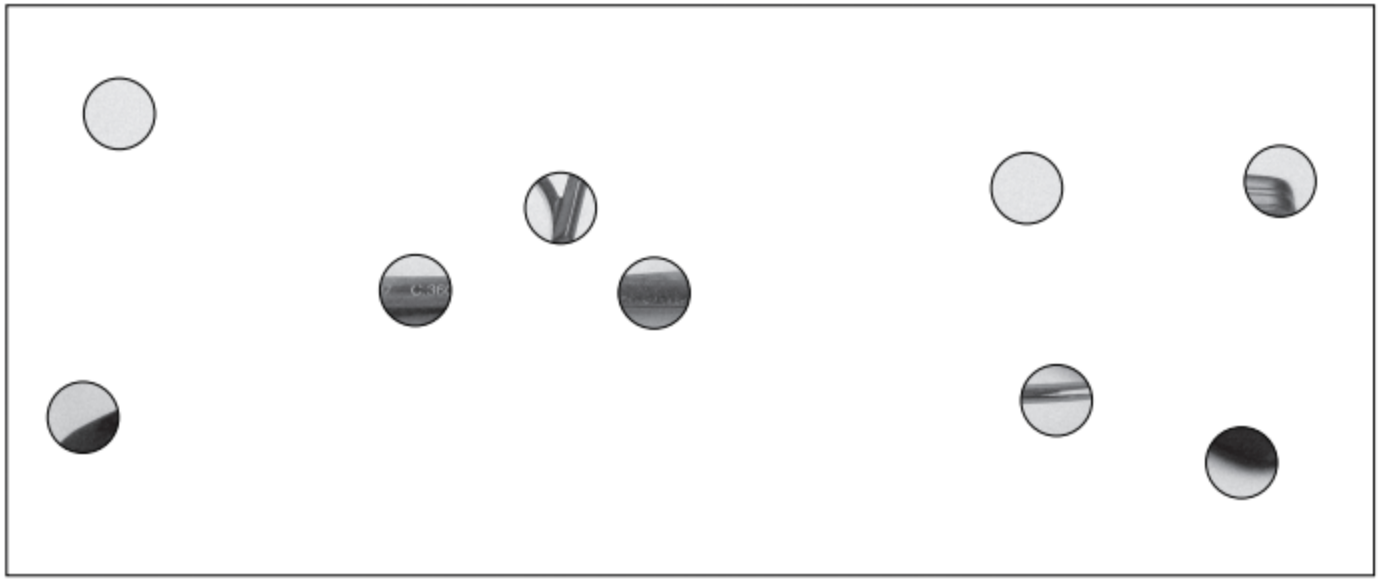
Vision is Hard

- A 2D picture may be produced by many different 3D scenes



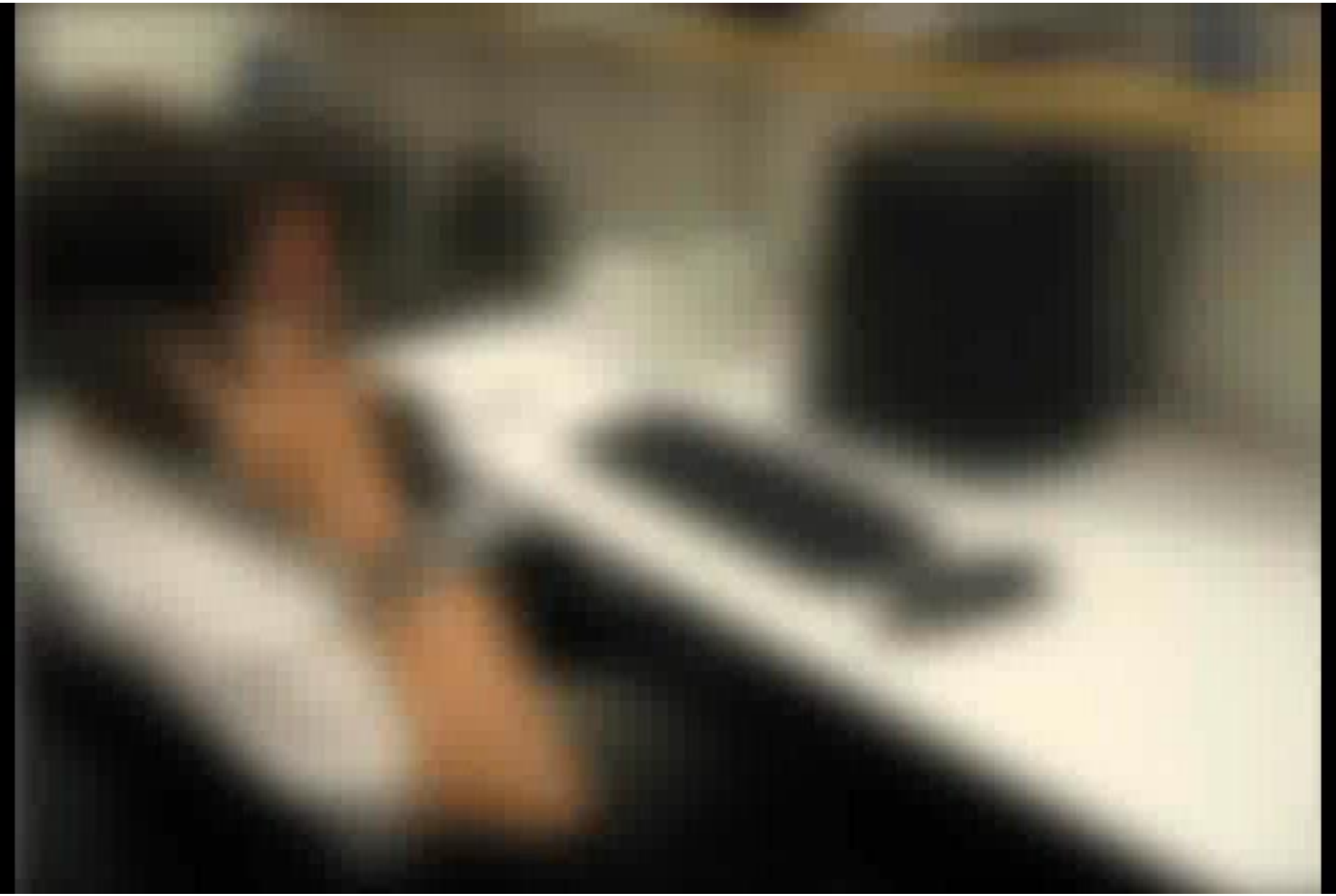


Vision is Hard



Vision is Hard







Why is Vision Hard?

- Loss of information due to projection from 3D to 2D
  - Infinite scenes could have generated a given image
- Image colors depend on surface properties, illumination, camera response function and interactions such as shadows
  - HVS very good at ignoring distractors
- Noise
  - sensor noise and nonlinearities, quantization
- Lots of data
- Conflicts among local and global cues
  - Illusions

The Horizon

- Not all hard to explain phenomena are unusual…



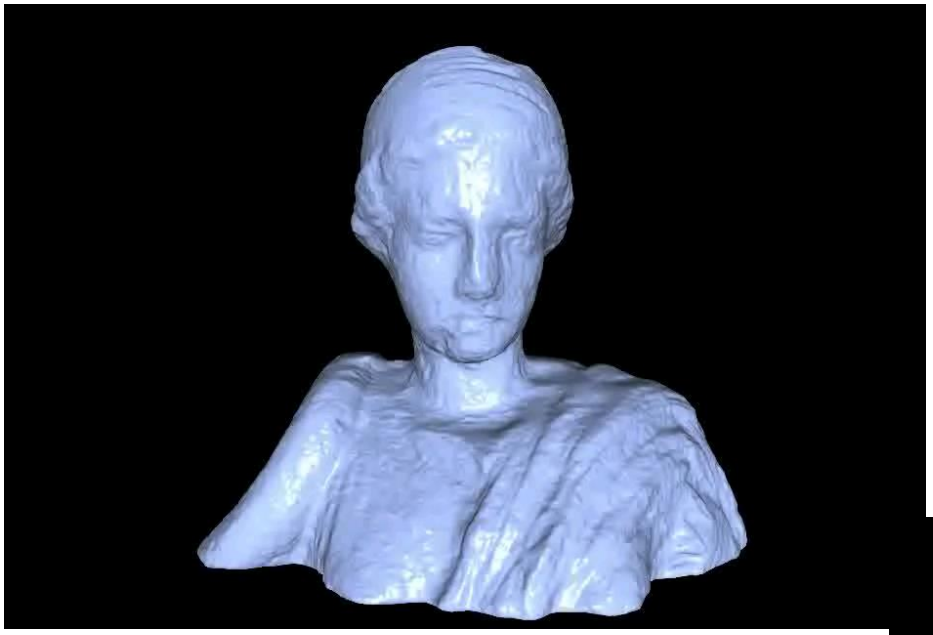
Vanishing Points

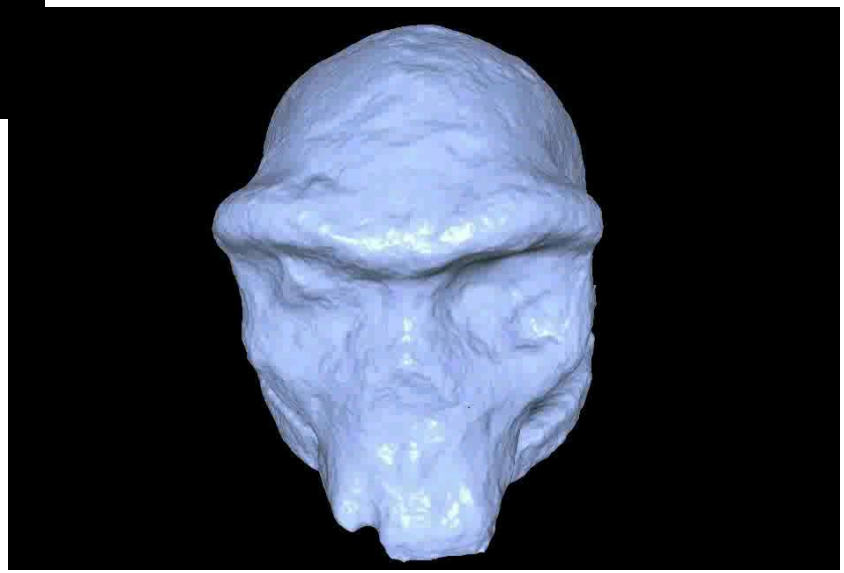


Why 3D Vision?

- Structure from Motion
  - Simultaneous Localization and Mapping
- 3D reconstruction
  - Dense mapping …
- 3D motion capture
- Medical applications
- Robotics and autonomous driving
  - Driver assistance

3D Models



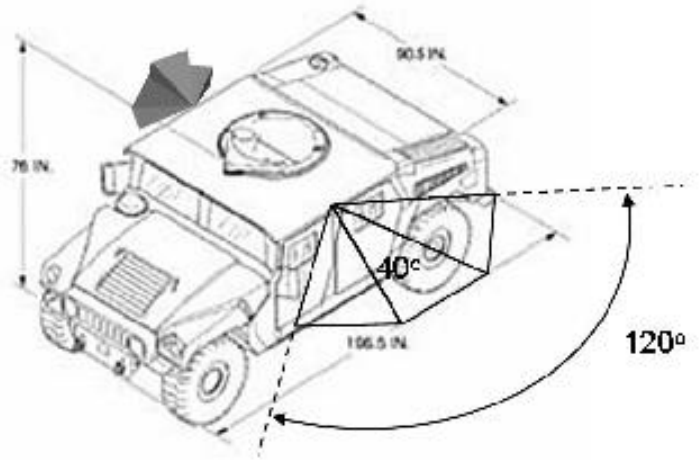


Real-Time Video-based 3D Reconstruction

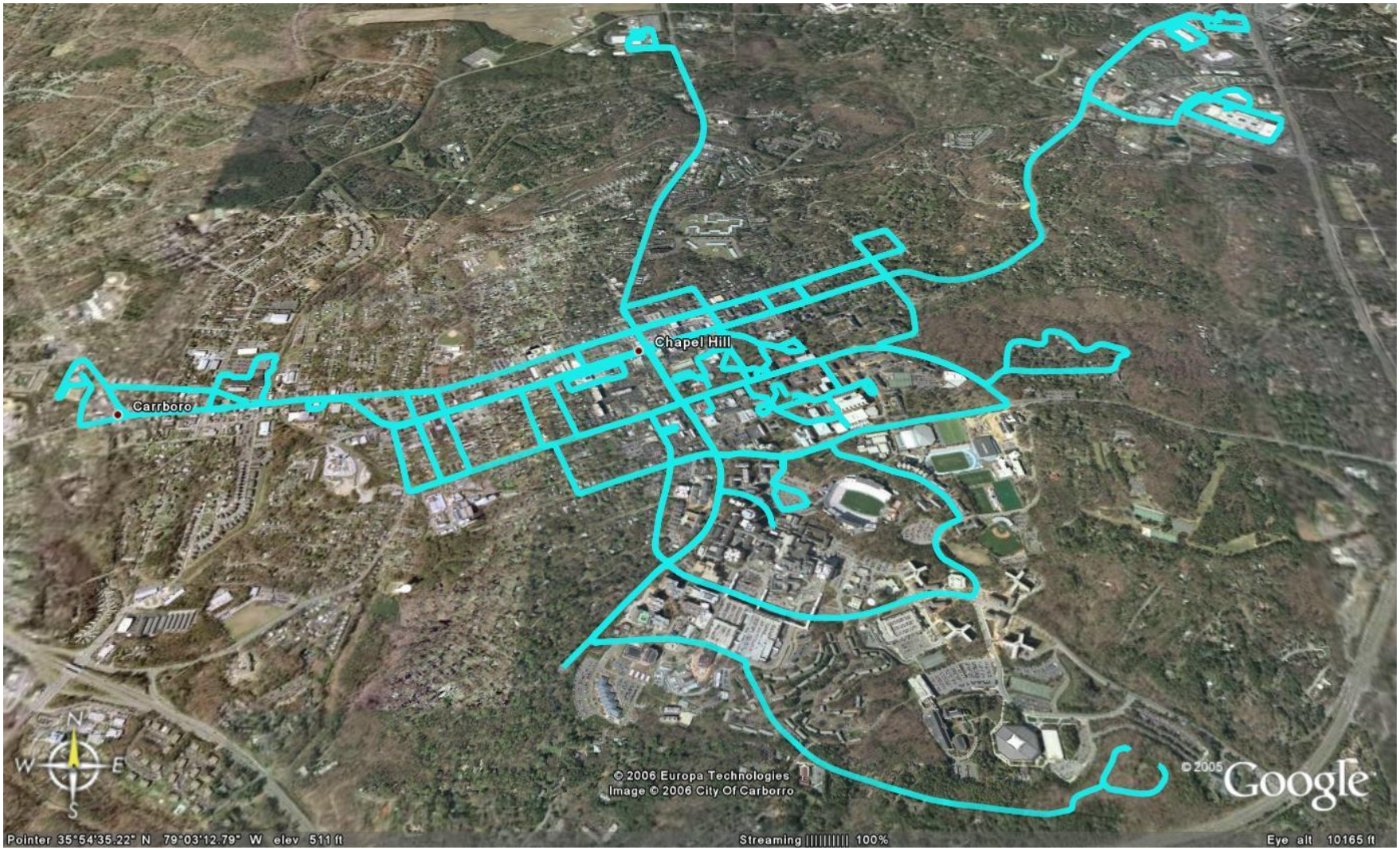
- Goal: real-time reconstruction of urban environments for visualization and training
- Platform:
  - 8 *non-overlapping* cameras
  - Differential GPS
  - Inertial Navigation System



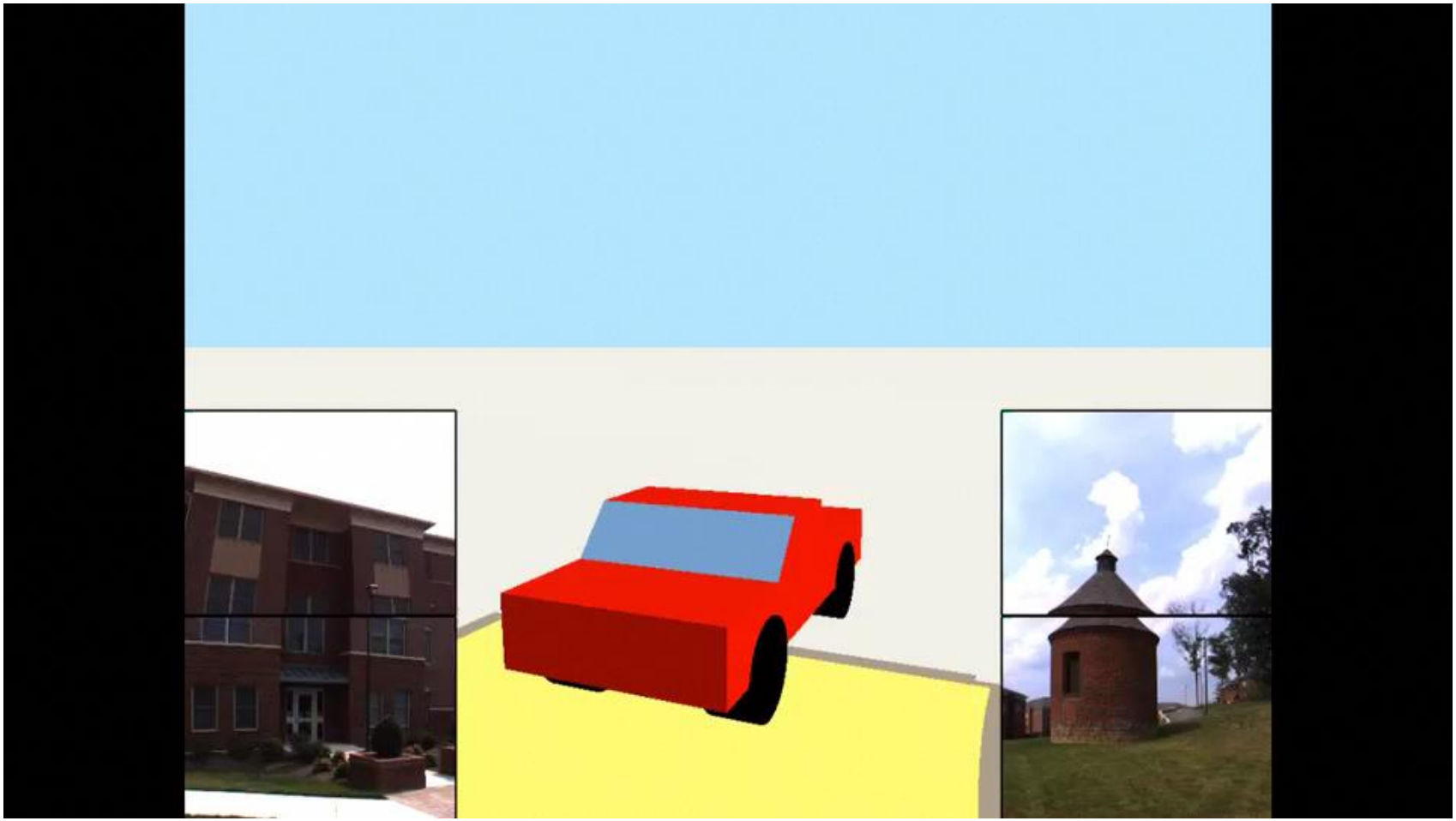




Data Collection

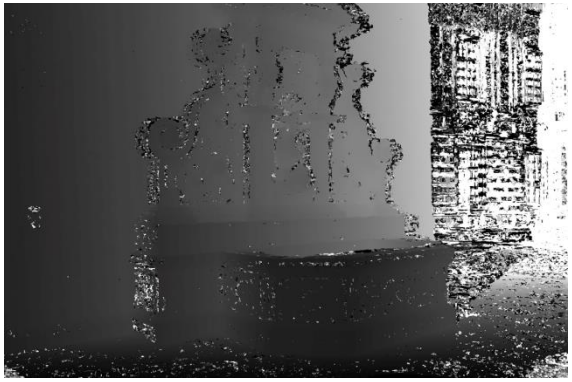
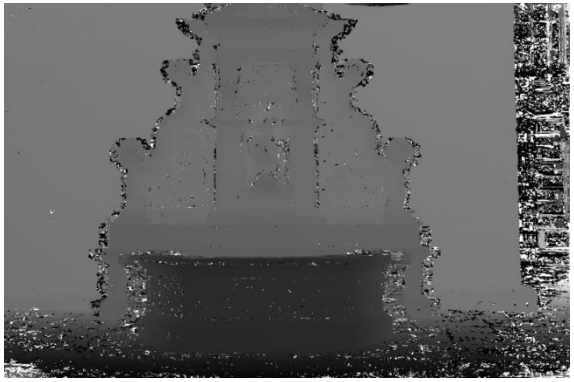
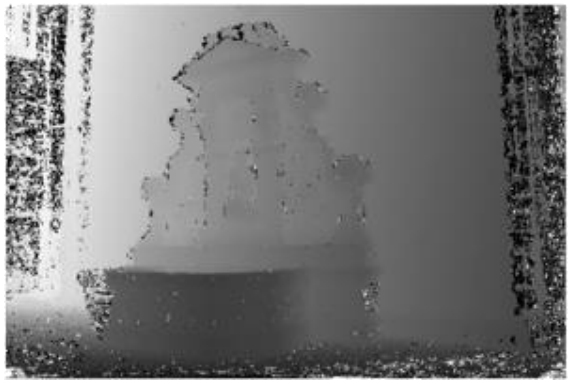


Results: Chapel Hill



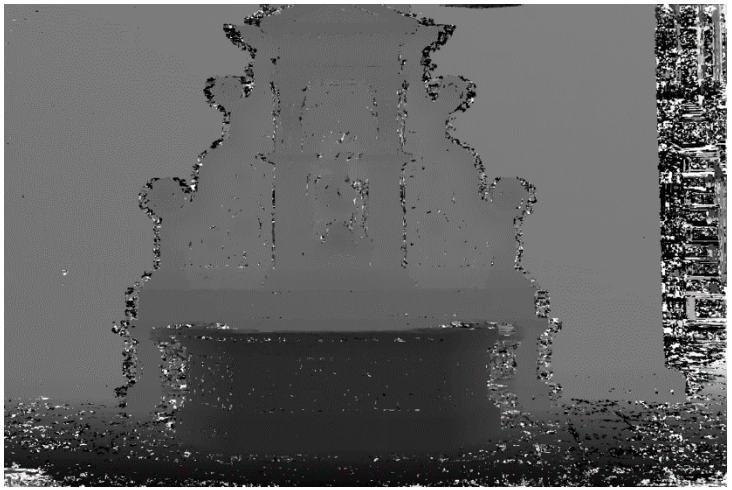
Depth Map Estimation



3 of 11 images and corresponding depth maps

Depth Map Fusion



Raw Depth Map



Fused Depth Map





Colored Point Clouds

Rome in a Day



The World in Six Days

Building the World in Six Days

CVPR 2015

Paper 964

Visual Turing Test (UW)

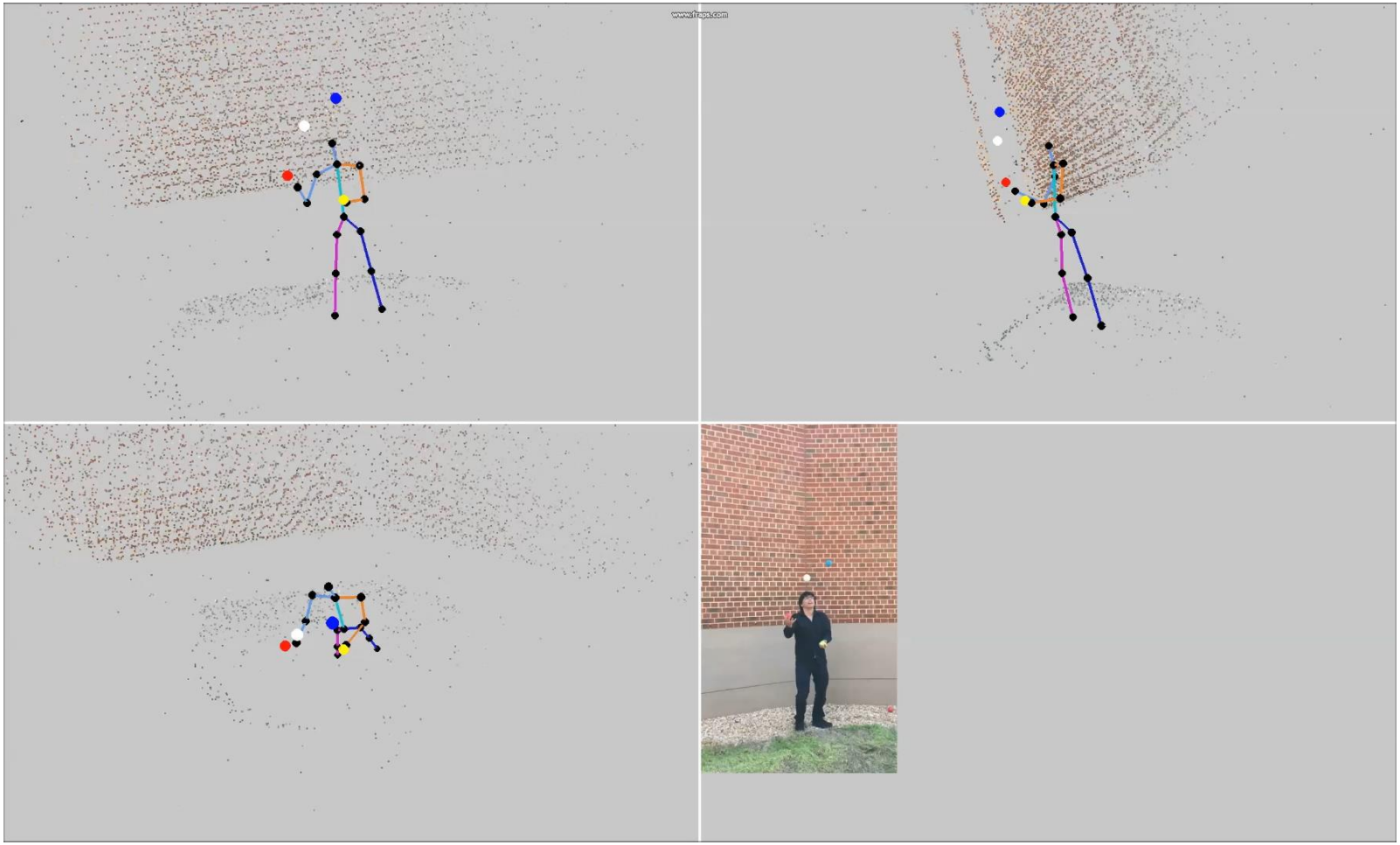


Shan, Adams, Curless, Furukawa and Seitz (2013)

Visual SLAM



Dynamic Reconstruction



The Internet as a Camera…

- Massively Distributed First-Person Capture

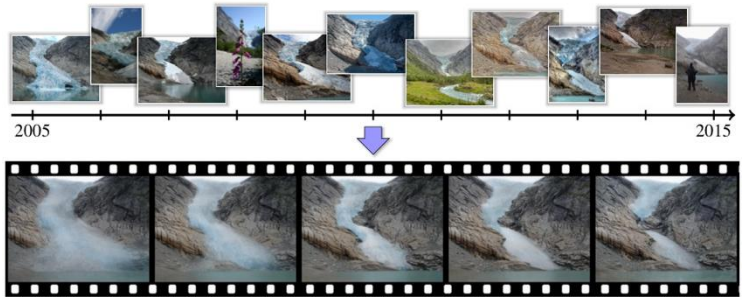



- Crowd-Sourced Motion Capture





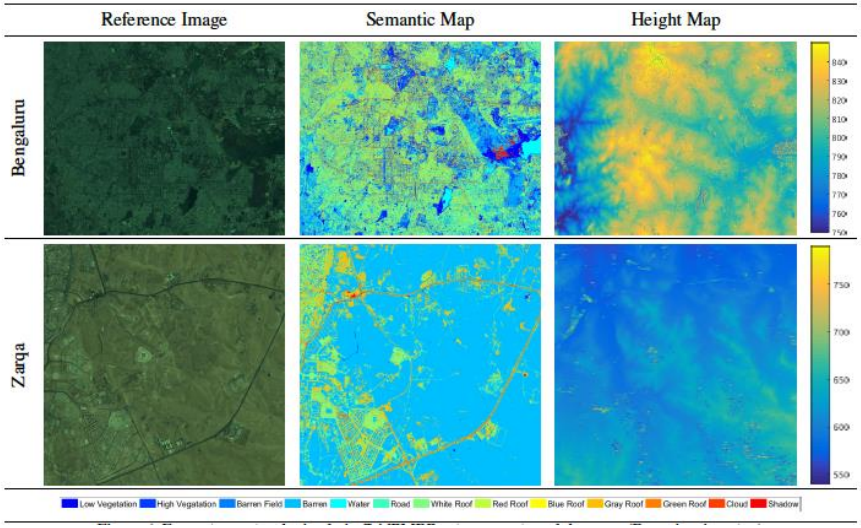
- Visual Time Machines

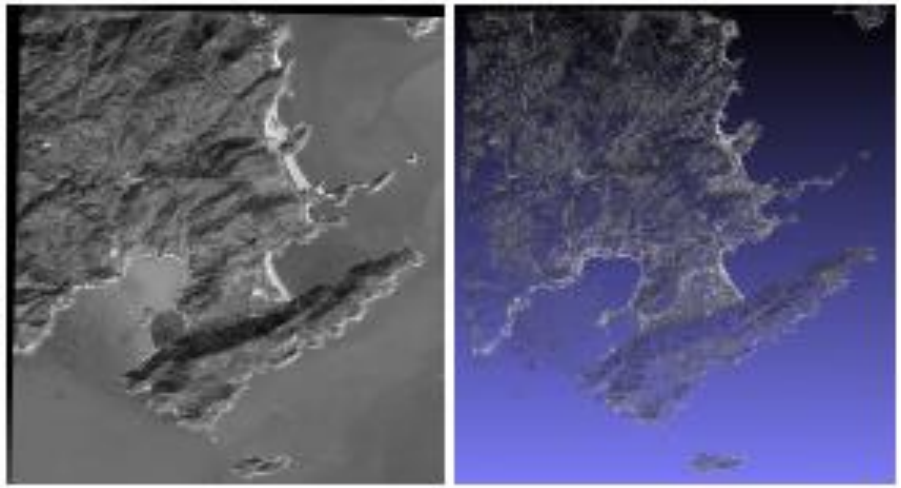


Martin-Brualla et al. SIGGRAPH 2015

Other Ongoing Efforts…

- Remote Sensing

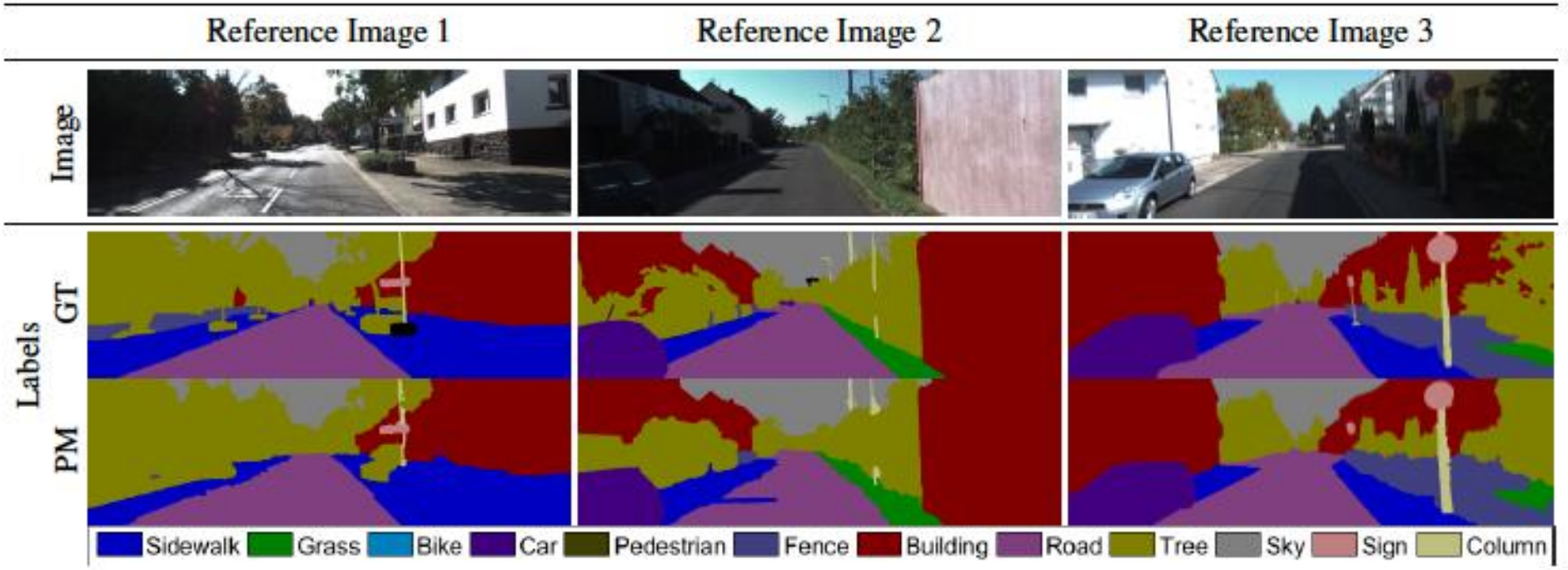




"Minimal Solvers for 3D Geometry from Satellite Imagery", ICCV 2015

Other Ongoing Efforts…

- Remote Sensing
- Autonomous Systems

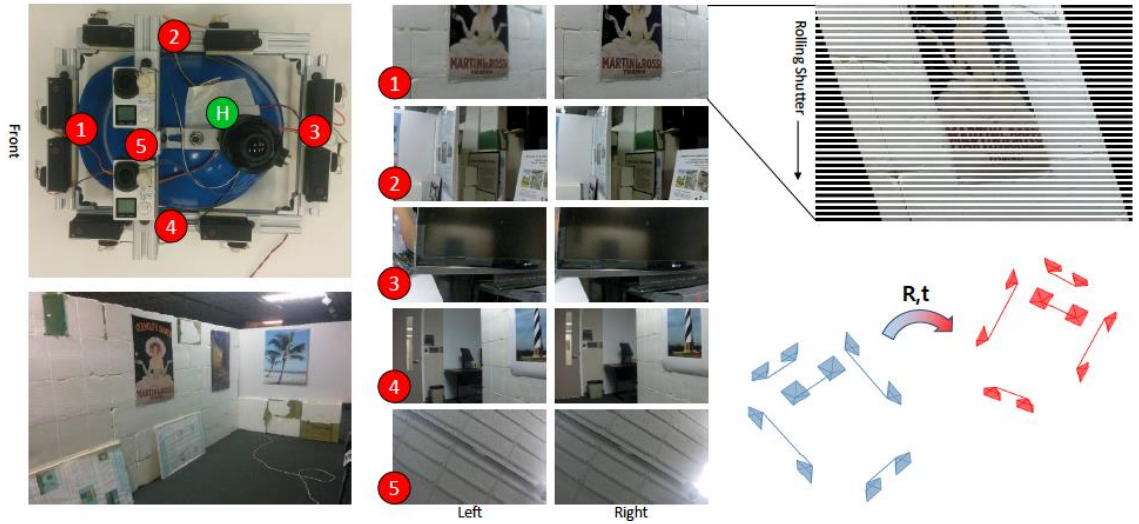


Combining Semantic Scene Priors and Haze Removal for Single Image Depth Estimation", WACV 2014

Other Ongoing Efforts…

- Remote Sensing
- Autonomous Systems
- Augmented and Virtual Reality





Rolling Shutter Tracker
Best paper award at ISMAR 2016
ISMAR-TVCG paper

Other Ongoing Efforts…

- Remote Sensing
- Autonomous Systems
- Augmented and Virtual Reality
- Geo-Localization

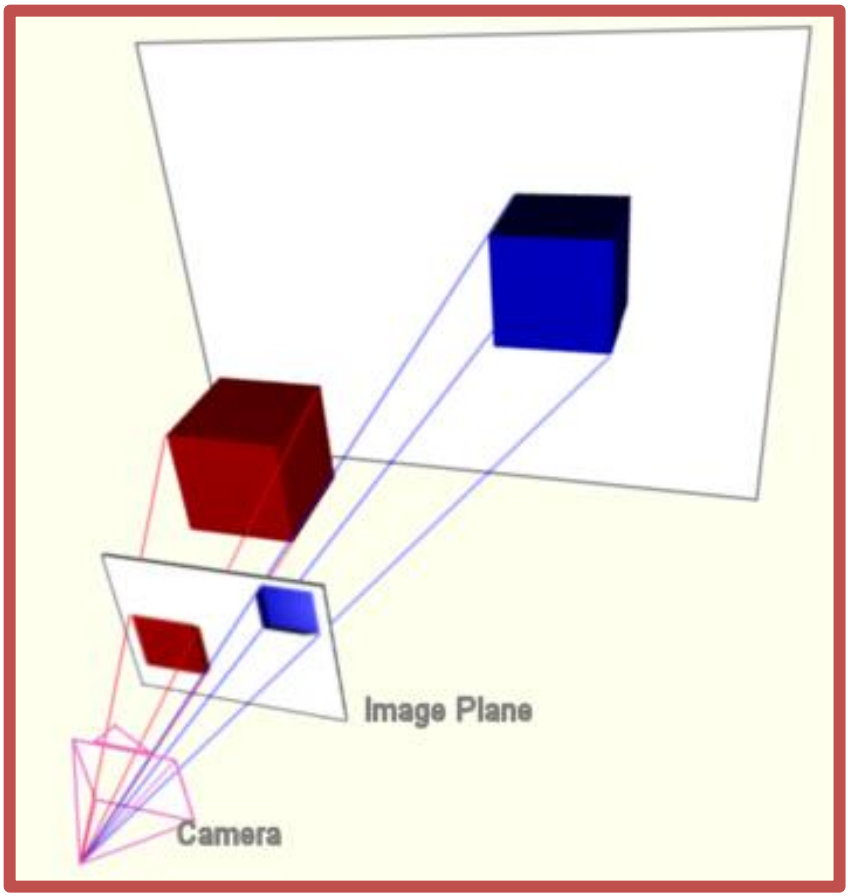


"Predicting Good Features for Image Geo-Localization Using Per-Bundle VLAD", ECCV 2015.

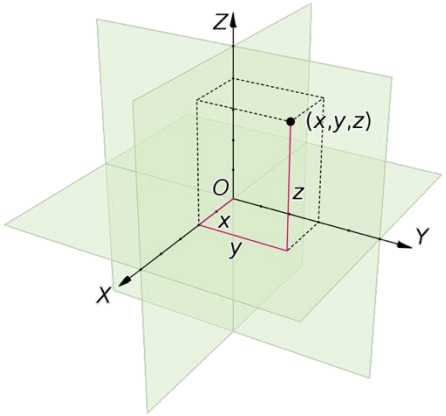
Introduction to Geometry

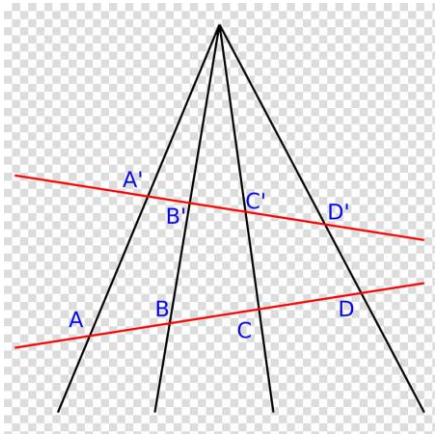
Based on slides by M. Pollefeys (ETH)
and D. Cappelleri (Purdue)

Pin-hole Camera Model



- Euclidean Geometry
- Projective Geometry
- Linear Algebra





2D Primitives

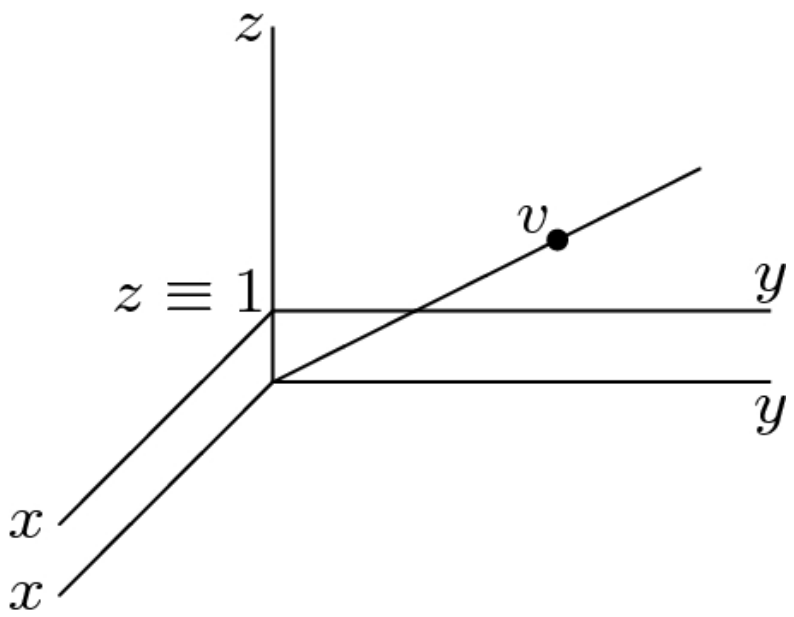
Points and Lines in 2D

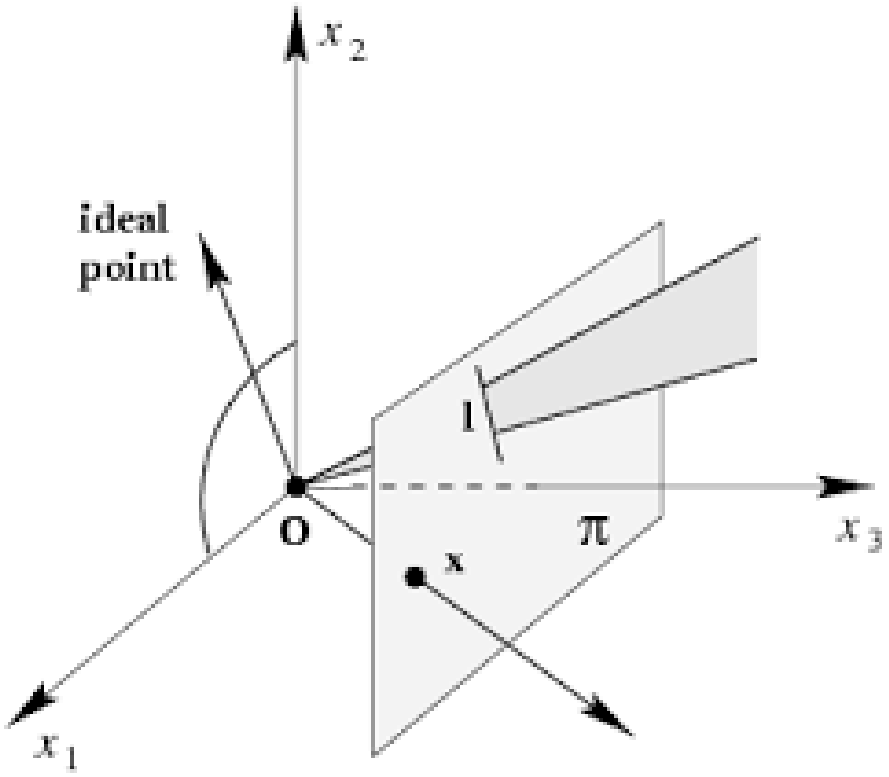
- A point (x, y) lies on a line (a, b, c) when:

  $ax+by+c = 0$ or $(a, b, c)(x, y, 1)^T = 0$

- Use homogeneous coordinates to represent points => add an extra coordinate
  - Note that scale is unimportant for determining incidence: k(x, y, 1) is also on the line
  - Homogeneous coordinates (x_1, x_2, x_3), but only two degrees of freedom
  - Equivalent to inhomogeneous coordinates (x, y)

Points and Lines in 2D





Points from Lines and Vice Versa

- The intersection of two lines l and l’ is given by: $l \times l'$

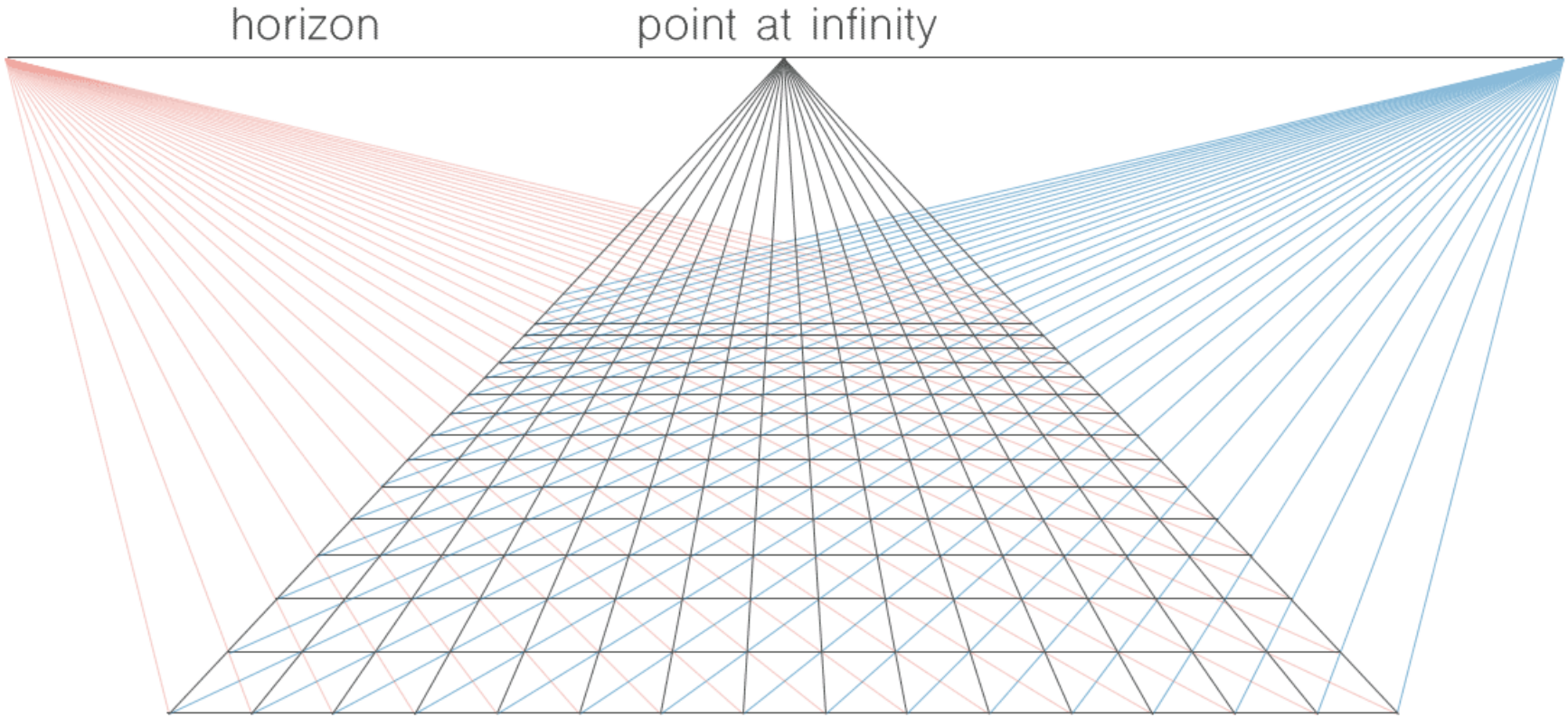
- The line connecting two points x and x’ is given by: $x \times x'$

$$\vec{C} = \vec{A} \times \vec{B} = \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \times \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} = \begin{pmatrix} A_y \cdot B_z - A_z \cdot B_y \\ A_z \cdot B_x - A_x \cdot B_z \\ A_x \cdot B_y - A_y \cdot B_x \end{pmatrix}$$

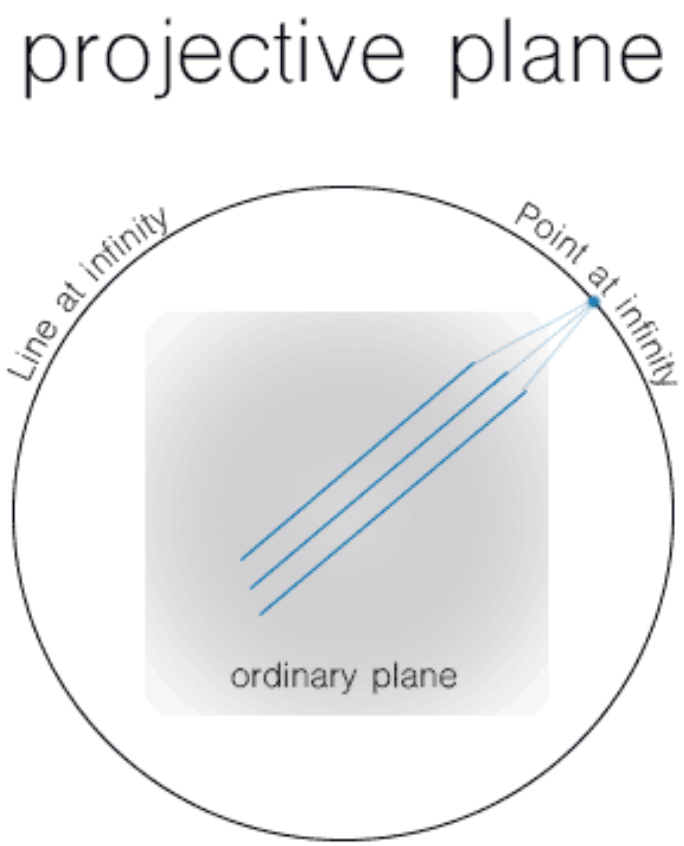
Ideal Points and the Line at Infinity



Ideal Points and the Line at Infinity



Ideal Points and the Line at Infinity



Ideal Points and the Line at Infinity

- Intersection of two parallel lines:
  - $l = (a, b, c)$ and $l' = (a, b, c')$
  - $l \times l' = (b, -a, 0)$

- Ideal points: $(x_1, x_2, 0)$
- Belong to the line at infinity $l = (0, 0, 1)$

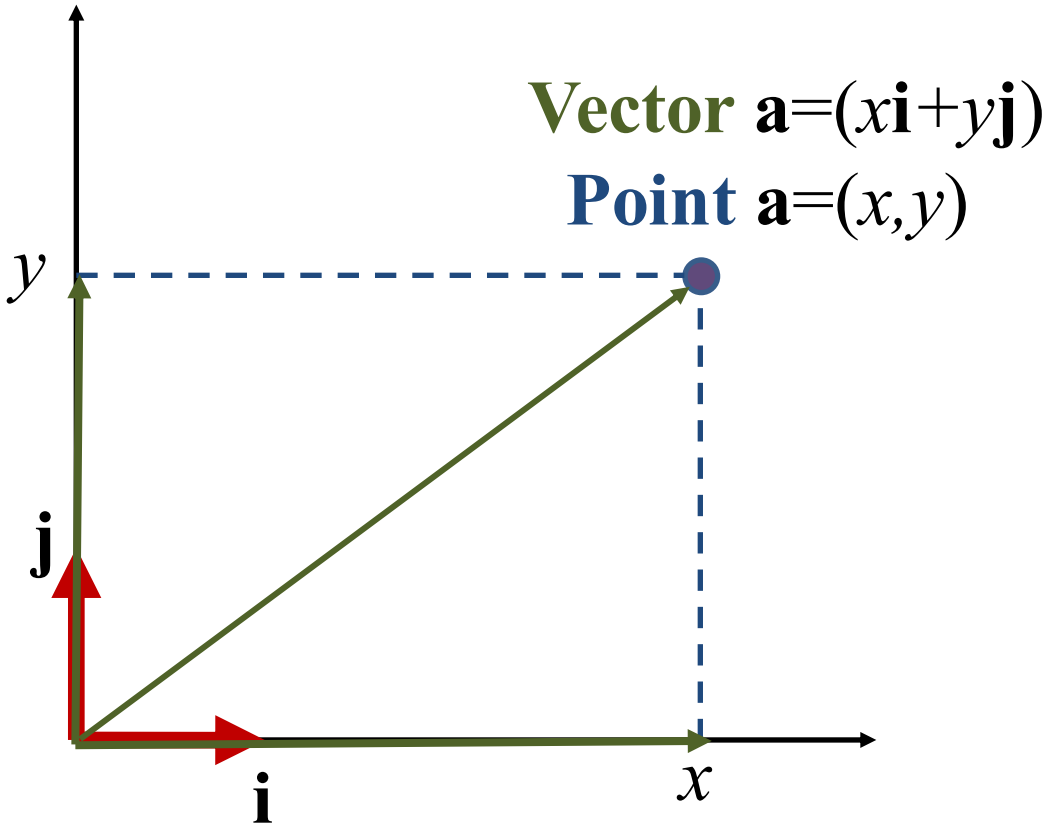
- $\mathbf{P}^2 = \mathbf{R}^3 - (0, 0, 0)$ (projective space)
  - In $\mathbf{P}^2$ there is no distinction between regular and ideal points

Rotation in 2D

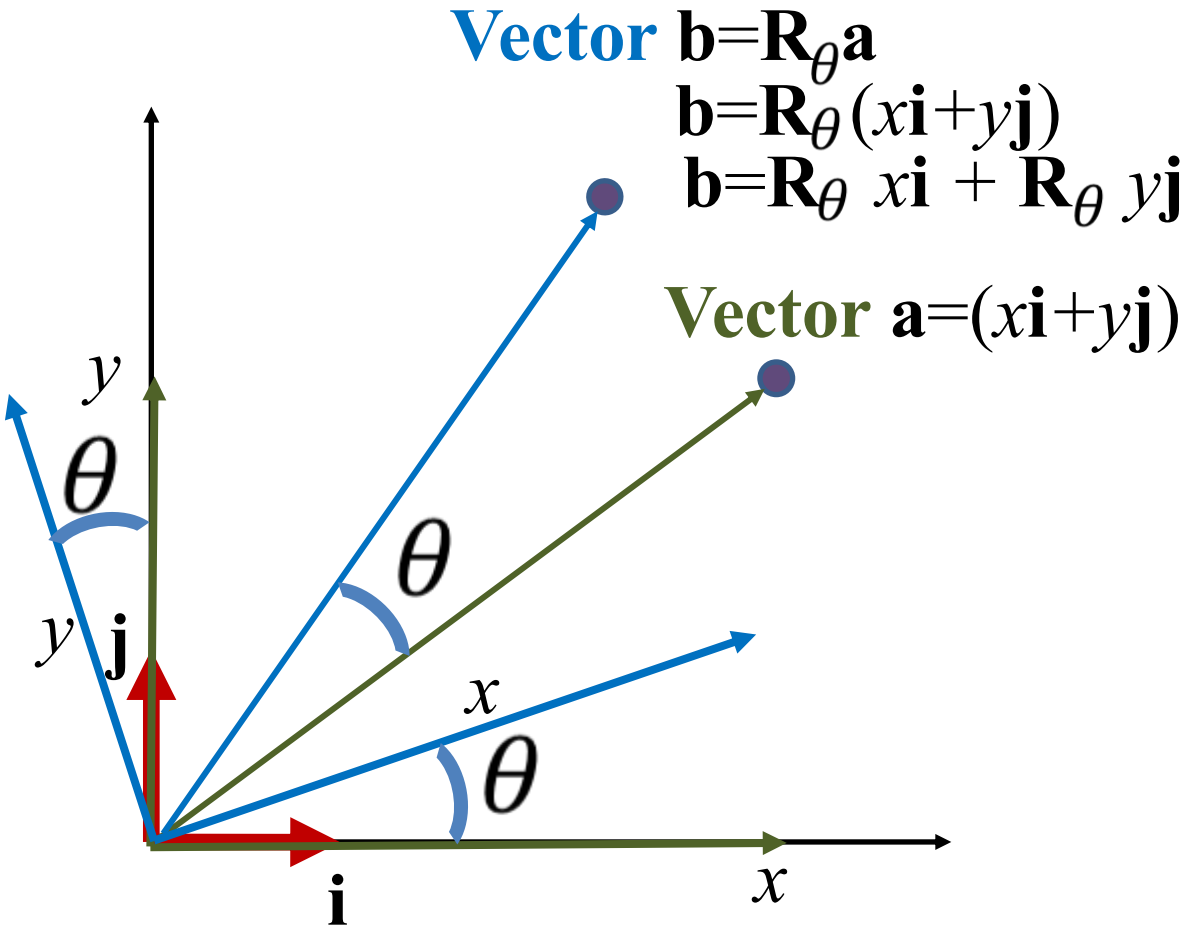
- Matrices are operators that transform vectors
  - 2D rotation matrix R=$\begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix}$

- In homogeneous coordinates $\begin{bmatrix} R & 0 \\ 0 & 1 \end{bmatrix}$

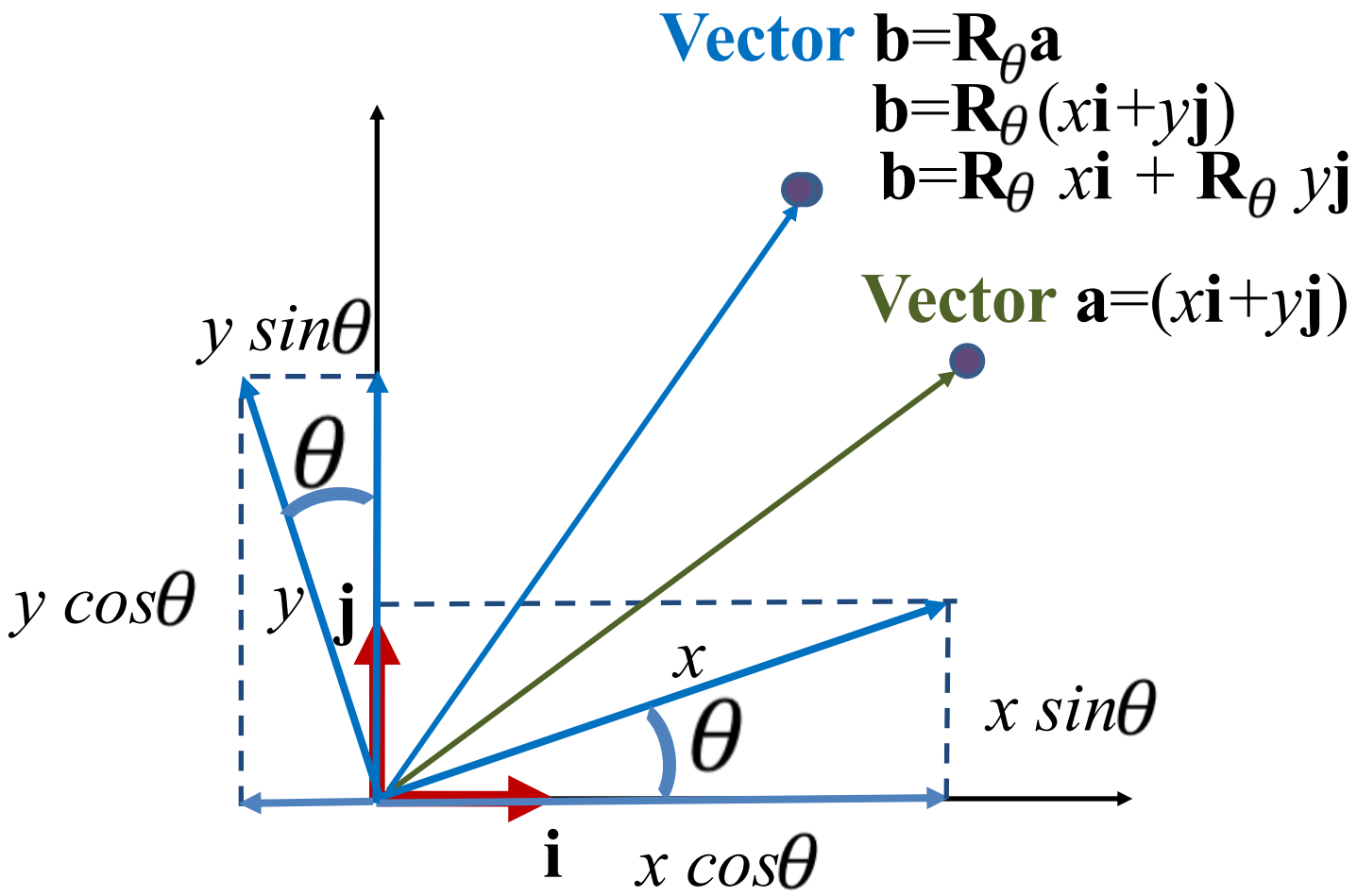
2D Rotations



2D Rotations



2D Rotations

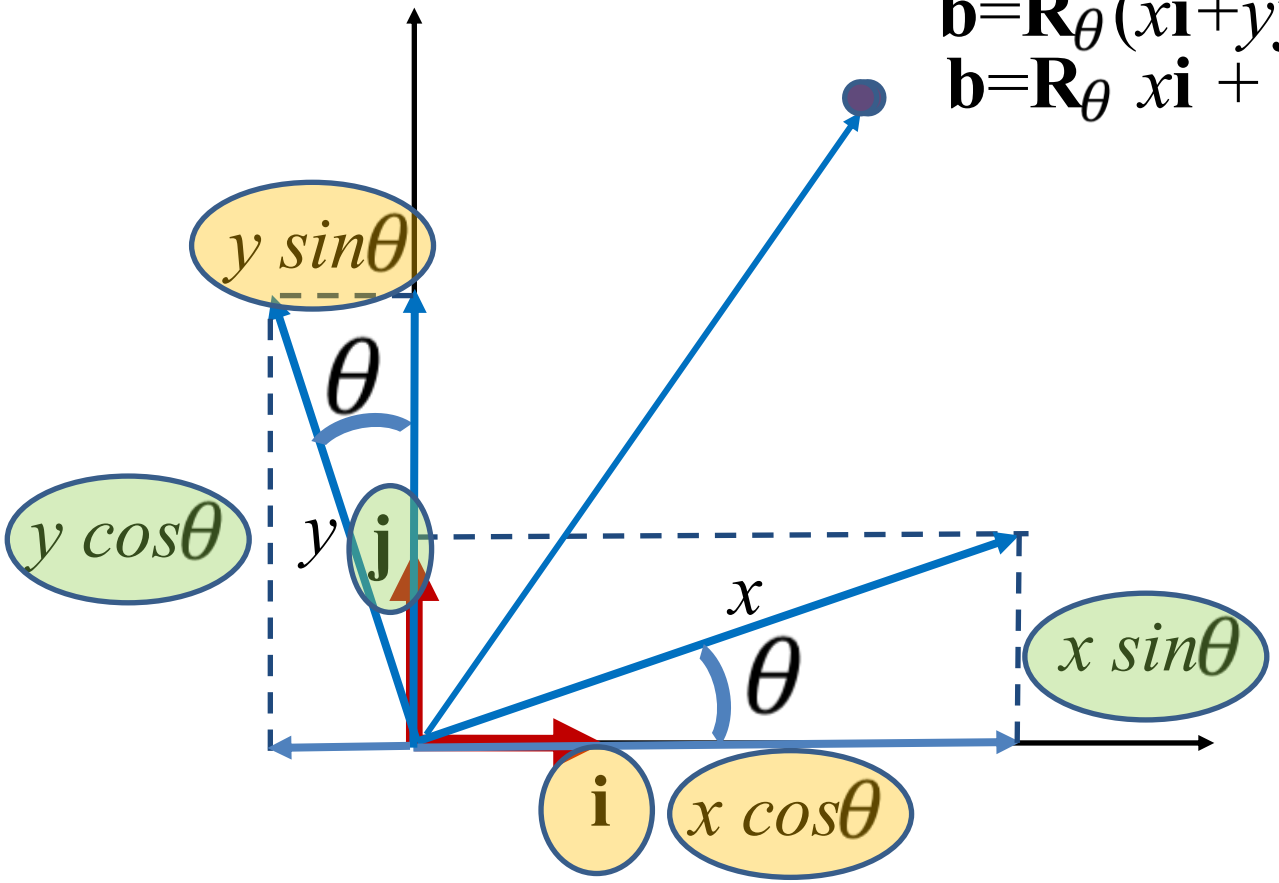


2D Rotations

Vector $\mathbf{b}=\mathbf{R}_\theta\mathbf{a}$

$\mathbf{b}=\mathbf{R}_\theta(x\mathbf{i}+y\mathbf{j})$

$\mathbf{b}=\mathbf{R}_\theta\, x\mathbf{i} + \mathbf{R}_\theta\, y\mathbf{j}$



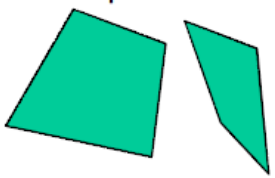
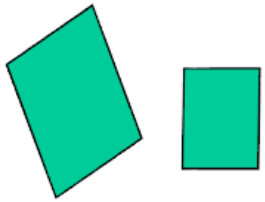
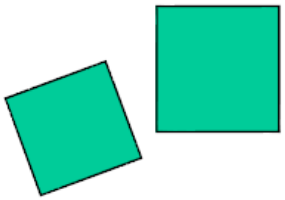
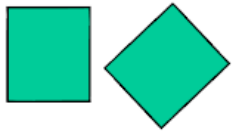
Hands-on: 2D Transformations

- How to translate a point in homogeneous and inhomogeneous coordinates?
- How to rotate a point around the origin?
- How to rotate a point around a center other than the origin?

$$R=\begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix}$$

$$\begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

Hierarchy of 2D Transformations

		transformed squares	invariants
Projective 8dof	$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$		Concurrency, collinearity, order of contact (intersection, tangency, inflection, etc.), cross ratio
Affine 6dof	$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Parallellism, ratio of areas, ratio of lengths on parallel lines (e.g midpoints), linear combinations of vectors (centroids). **The line at infinity l_∞**
Similarity 4dof	$\begin{bmatrix} sr_{11} & sr_{12} & t_x \\ sr_{21} & sr_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Ratios of lengths, angles. **The circular points I,J**
Euclidean 3dof	$\begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		lengths, areas.

Transformation of Points and Lines

Point transformation

$$\mathbf{x}' = \mathbf{H}\,\mathbf{x}$$

Line transformation

$$\mathbf{l}' = \mathbf{H}^{-T}\,\mathbf{l}$$

3D Primitives

3D points

3D point

$(X, Y, Z)^{\mathsf{T}}$ in $\mathbf{R}^3$

$\mathrm{X} = (X_1, X_2, X_3, X_4)^{\mathsf{T}}$ in $\mathbf{P}^3$

$$\mathrm{X} = \left(\frac{X_1}{X_4}, \frac{X_2}{X_4}, \frac{X_3}{X_4}, 1\right)^{\mathsf{T}} = (X, Y, Z, 1)^{\mathsf{T}} \quad (X_4 \neq 0)$$

projective transformation

$\mathrm{X}' = \mathbf{H}\,\mathrm{X}$ (4x4-1=15 dof)

Planes

3D plane

$$\pi_1 X + \pi_2 Y + \pi_3 Z + \pi_4 = 0$$

$$\pi_1 X_1 + \pi_2 X_2 + \pi_3 X_3 + \pi_4 X_4 = 0$$

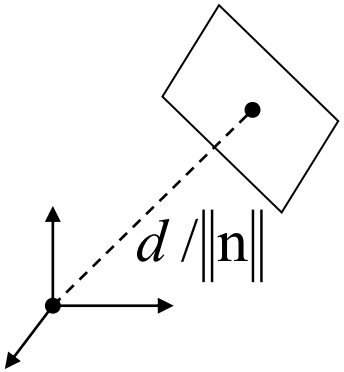
$$\pi^{\mathsf{T}} \mathrm{X} = 0$$

Transformation

$$\mathrm{X}' = \mathbf{H}\,\mathrm{X}$$
$$\pi' = \mathbf{H}^{-\mathsf{T}}\,\pi$$

Euclidean representation

$$\mathrm{n}.\widetilde{\mathrm{X}} + d = 0 \qquad \mathrm{n} = (\pi_1, \pi_2, \pi_3)^{\mathsf{T}} \qquad \widetilde{\mathrm{X}} = (X, Y, Z)^{\mathsf{T}}$$
$$\pi_4 = d \qquad X_4 = 1$$



Planes from points

Solve π from $X_1^T\pi = 0, X_2^T\pi = 0$ and $X_3^T\pi = 0$

$$\begin{bmatrix} X_1^T \\ X_2^T \\ X_3^T \end{bmatrix} \pi = 0$$ (solve as right nullspace of π)

$$\begin{bmatrix} X_1^T \\ X_2^T \\ X_3^T \end{bmatrix}$$

Planes from points

Solve π from $X_1^T\pi = 0, X_2^T\pi = 0$ and $X_3^T\pi = 0$

$$\begin{bmatrix} X_1^T \\ X_2^T \\ X_3^T \end{bmatrix} \pi = 0 \quad \text{(solve as right nullspace of } \pi \text{)} \qquad \begin{bmatrix} X_1^T \\ X_2^T \\ X_3^T \end{bmatrix}$$

Or implicitly from coplanarity condition

$$\det \begin{bmatrix} X_1 & (X_1)_x & (X_2)_x & (X_3)_x \\ X_2 & (X_1)_y & (X_2)_y & (X_3)_y \\ X_3 & (X_1)_z & (X_2)_z & (X_3)_z \\ X_4 & (X_1)_w & (X_2)_w & (X_3)_w \end{bmatrix} = 0$$

$$X_1 D_{234} - X_2 D_{134} + X_3 D_{124} - X_4 D_{123} = 0$$

$$\pi = \left(D_{234}, -D_{134}, D_{124}, -D_{123}\right)^T$$

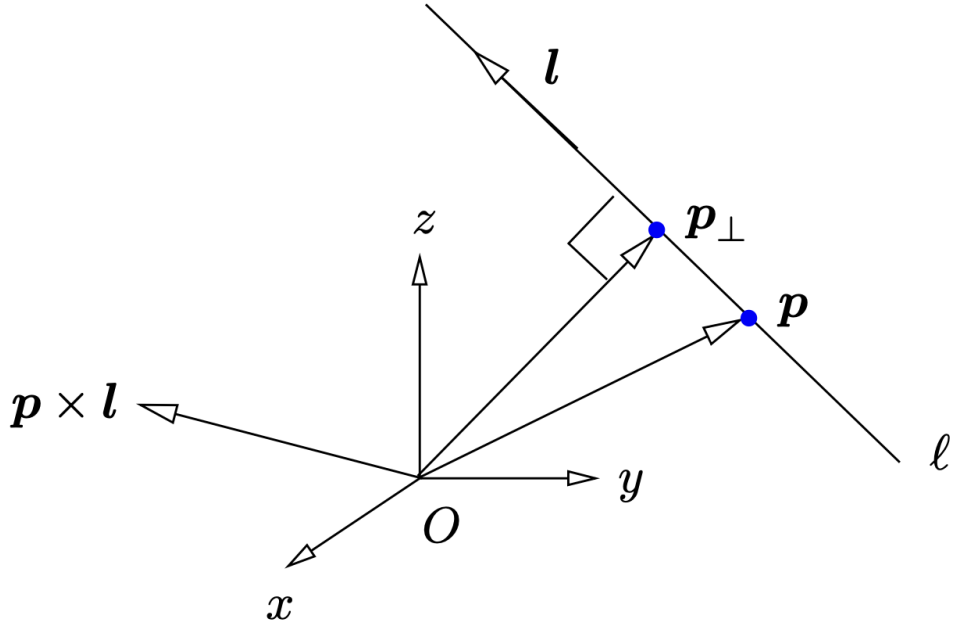
Points from planes

Solve $\mathbf{X}$ from $\pi_1^T \mathbf{X} = 0$, $\pi_2^T \mathbf{X} = 0$ and $\pi_3^T \mathbf{X} = 0$

$$\begin{bmatrix} \pi_1^T \\ \pi_2^T \\ \pi_3^T \end{bmatrix} \mathbf{X} = 0$$ (solve as right nullspace of $\mathbf{X}$) $$\begin{bmatrix} \pi_1^T \\ \pi_2^T \\ \pi_3^T \end{bmatrix}$$

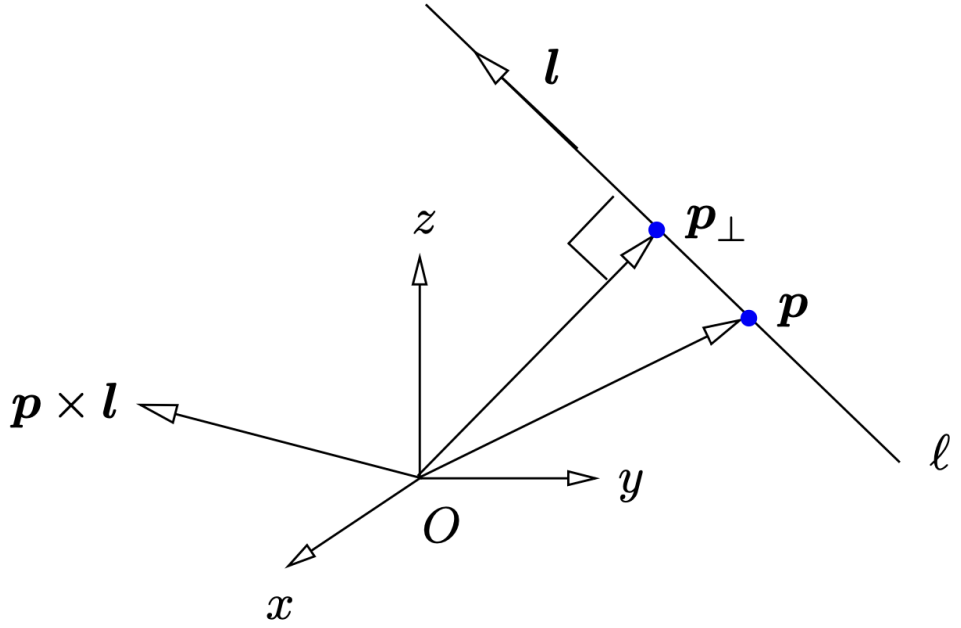
Lines are complicated (they live in P^5)…

Plucker Lines



Plücker coordinates $(\boldsymbol{l}, \boldsymbol{p} \times \boldsymbol{l})$ of a line. Here $\boldsymbol{p}_\perp$ is the closest point on the line to the origin O.

Plucker Lines



Plücker coordinates $(\boldsymbol{l}, \boldsymbol{p} \times \boldsymbol{l})$ of a line. Here $\boldsymbol{p}_\perp$ is the closest point on the line to the origin O.

Reciprocal Product

$$(\hat{\boldsymbol{l}}_1, \boldsymbol{m}_1) * (\hat{\boldsymbol{l}}_2, \boldsymbol{m}_2) \stackrel{\text{def}}{=} \hat{\boldsymbol{l}}_1 \cdot \boldsymbol{m}_2 + \hat{\boldsymbol{l}}_2 \cdot \boldsymbol{m}_1.$$

Distance Between Lines

Two lines ℓ_1 and ℓ_2, in Plücker coordinates $(\boldsymbol{l}_1, \boldsymbol{m}_1)$ and $(\boldsymbol{l}_2, \boldsymbol{m}_2)$, have the distance

$$d = \begin{cases} \dfrac{|(\boldsymbol{l}_1, \boldsymbol{m}_1) * (\boldsymbol{l}_2, \boldsymbol{m}_2)|}{\|\boldsymbol{l}_1 \times \boldsymbol{l}_2\|}, & \text{if } \boldsymbol{l}_1 \times \boldsymbol{l}_2 \neq 0; \\ \dfrac{\|\boldsymbol{l}_1 \times (\boldsymbol{m}_1 - \boldsymbol{m}_2/s)\|}{\|\boldsymbol{l}_1\|^2}, & \text{otherwise, where } \boldsymbol{l}_2 = s\boldsymbol{l}_1 \text{ for some } s \neq 0. \end{cases} \tag{10}$$

Rotations

- Rotation matrices around the 3 axes

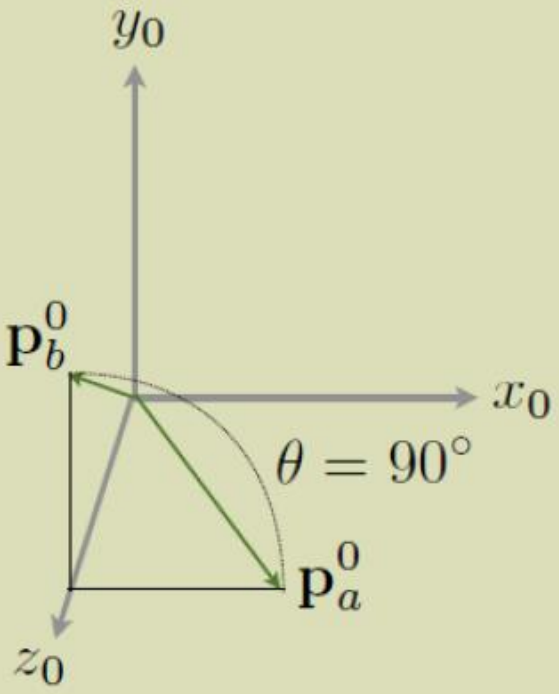
=> What is the inverse of a rotation matrix?

$$\mathbf{R}_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$\mathbf{R}_{y,\theta} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$\mathbf{R}_{z,\theta} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation Example



The rotation matrix can be used to perform arbitrary rotations on vectors

$$\mathbf{v}_p^0 = \mathbf{R}_1^0 \, \mathbf{v}_p^1$$

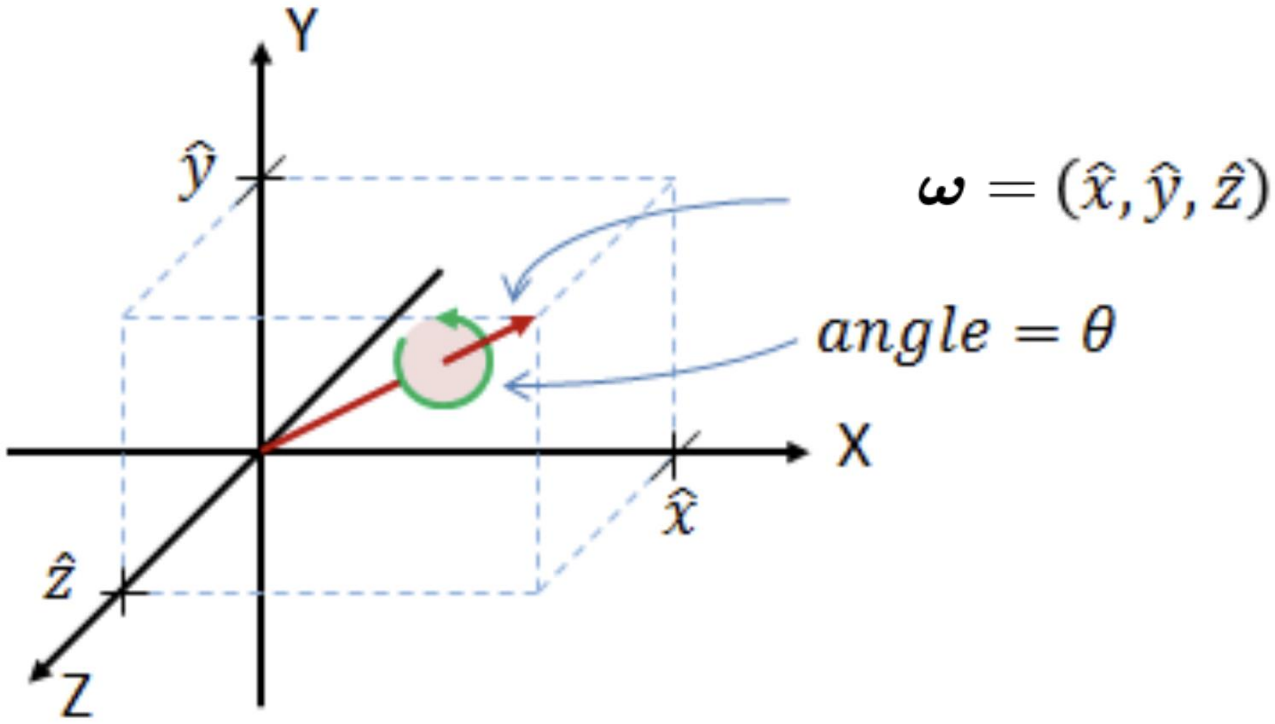
$$\mathbf{p}_a^0 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\mathbf{p}_b^0 = \mathbf{R}_{z,\theta} \, \mathbf{p}_a^0 = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Parameterization of Rotations

- In 3D, the 9-element rotation matrix has 3 DOF
- Several methods exist for representing a 3D rotation
  - Euler angles
  - Pitch, Roll, Yaw angles
  - Axis/Angle representation
  - Quaternions

Axis-angle



$$\theta = \arccos\left(\frac{\mathrm{Tr}(R) - 1}{2}\right)$$

$$\boldsymbol{\omega} = \frac{1}{2\sin\theta}\begin{bmatrix} R_{32} - R_{23} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{bmatrix}$$

Quaternions

Quaternion Representation

$$\boldsymbol{q} = (q_0, q_1, q_2, q_3)$$

where:

$$q_0 = \cos\left(\frac{\theta}{2}\right)$$

$$q_1 = \hat{x}\sin\left(\frac{\theta}{2}\right)$$

$$q_2 = \hat{y}\sin\left(\frac{\theta}{2}\right)$$

$$q_3 = \hat{z}\sin\left(\frac{\theta}{2}\right)$$

Quaternion Use

Transform point into quaternion

$$\boldsymbol{p} = (p_0, p_1, p_2, p_3) = (\,0, x, y, z\,)$$

Rotation is now Quaternion Multiplication

$$\boldsymbol{p'} = \boldsymbol{q}^{-1}\,\boldsymbol{pq}$$

Where multiplication is defined as

$$q1 = (w1, \mathbf{v}1), \quad q2 = (w2, \mathbf{v}2)$$

$$q1q2 = (\,w1w2 - \mathbf{v}1 \cdot \mathbf{v}2,\; w1\mathbf{v}2 + w2\mathbf{v}1 + \mathbf{v}1 \times \mathbf{v}2\,)$$

And quaternion inversions is defined as

$$\boldsymbol{q}^{-1} = (\,q_0, -q_1, -q_2, -q_3\,)$$

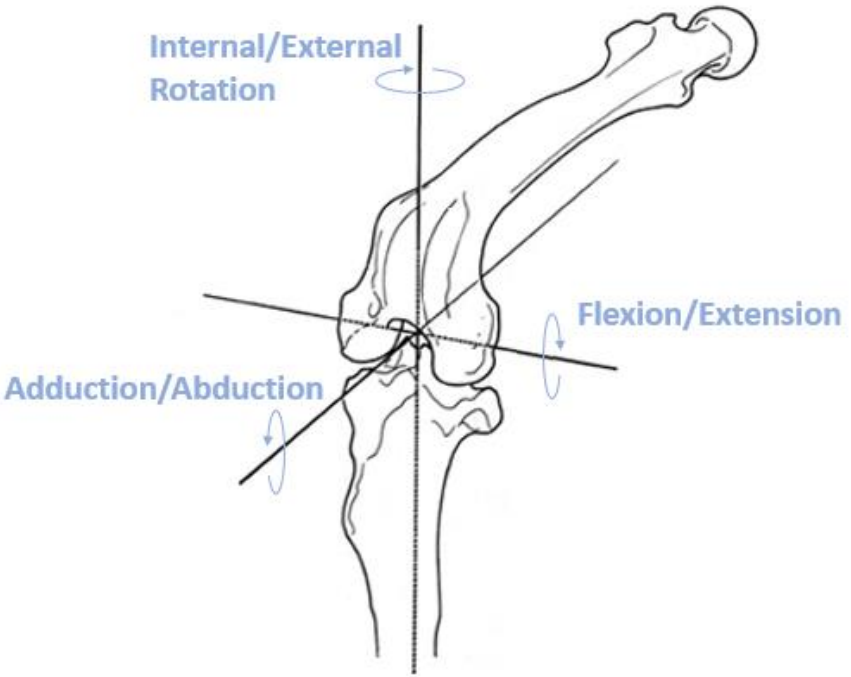
Rotation are compositional

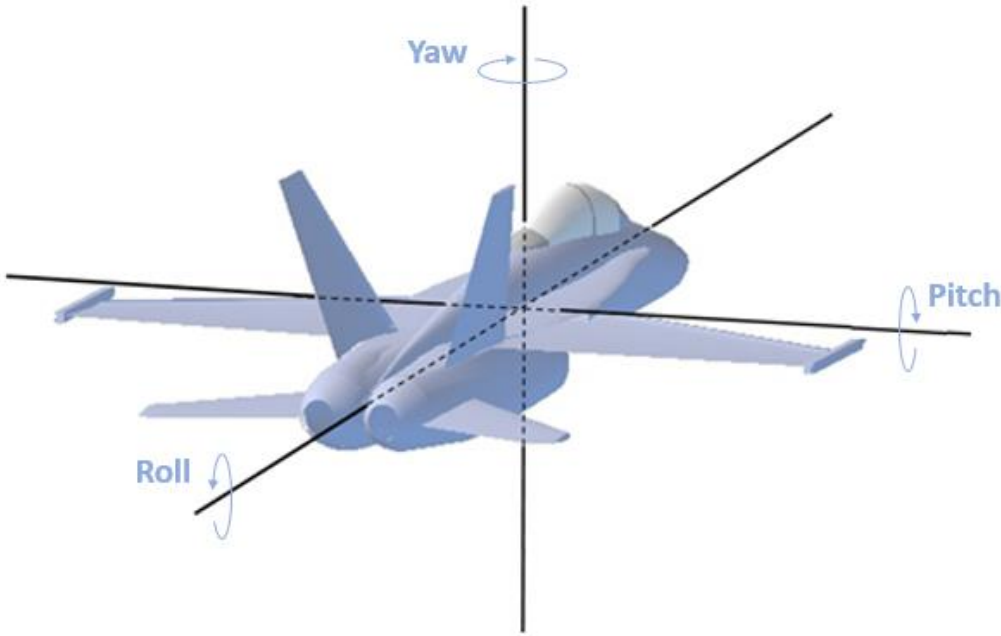
- Three consecutive elemental rotations (rotations about the axes of a coordinate system) are sufficient to reach any target frame.
- But which coordinate systems do we use?
- “Global” vs “Body” Coordinate Systems

BodyXYZ	BodyYZX	BodyZXY	BodyXZY	BodyYXZ	BodyZYX
BodyXYX	BodyXZX	BodyYXY	BodyYZY	BodyZXZ	BodyZYZ
GlobalXYZ	GlobalYZX	GlobalZXY	GlobalXZY	GlobalYXZ	GlobalZYX
GlobalXYX	GlobalXZX	GlobalYXY	GlobalYZY	GlobalZXZ	GlobalZYZ

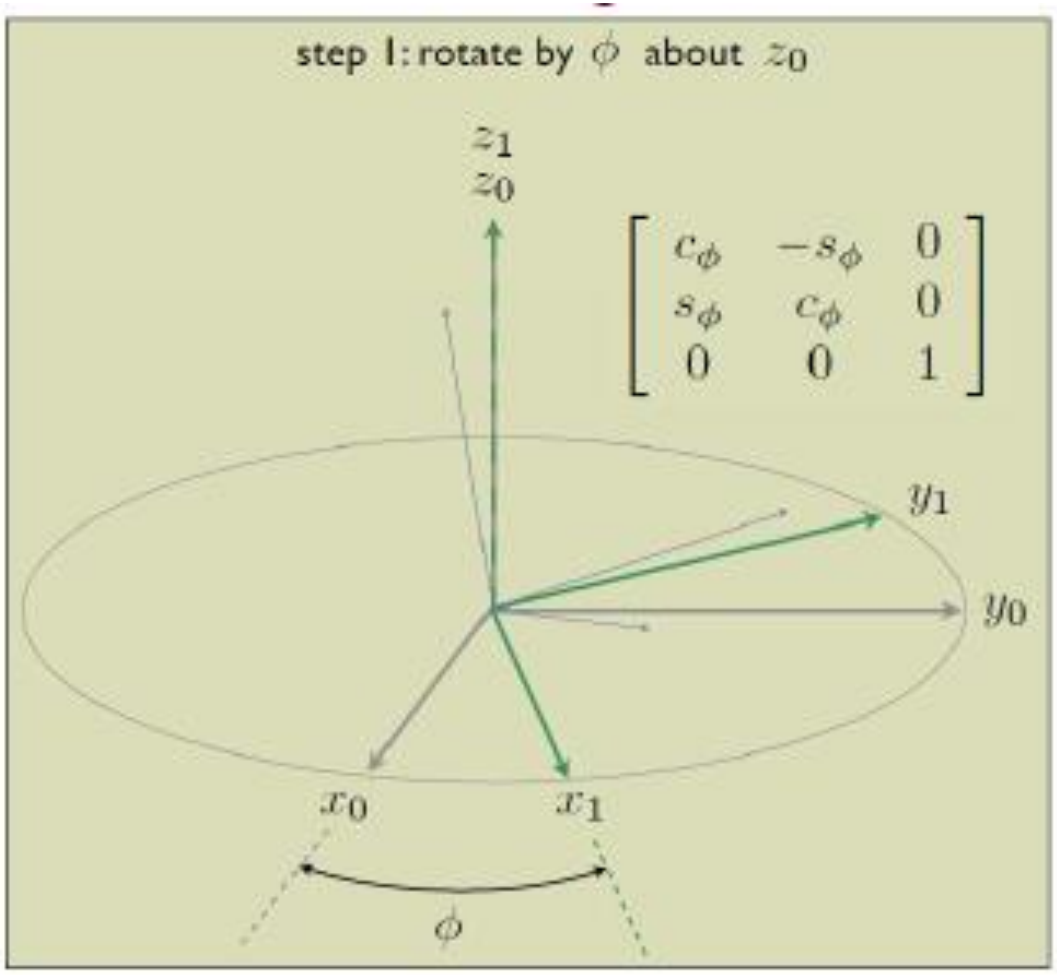
Euler Angles

- Define in Object-based coordinate systems

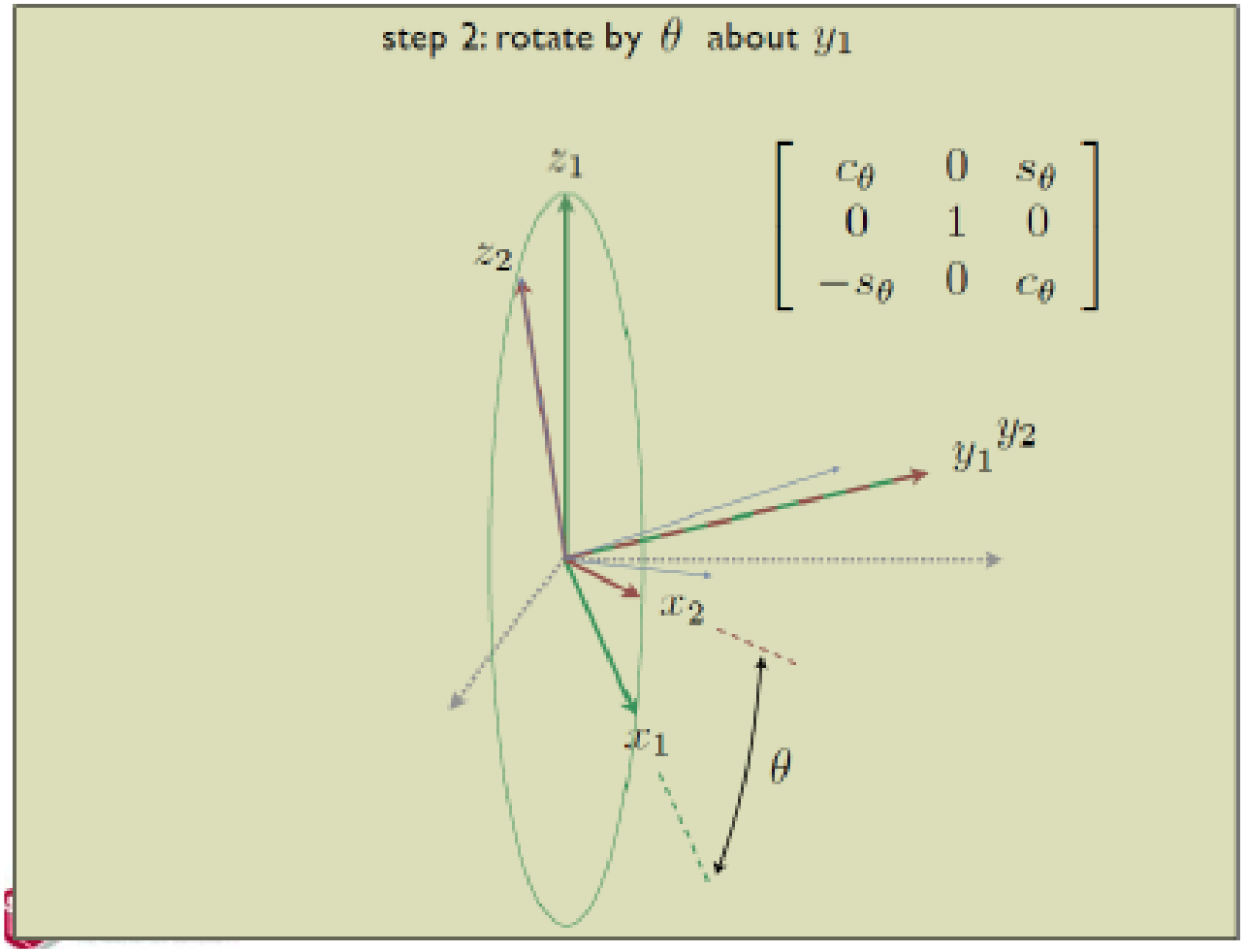




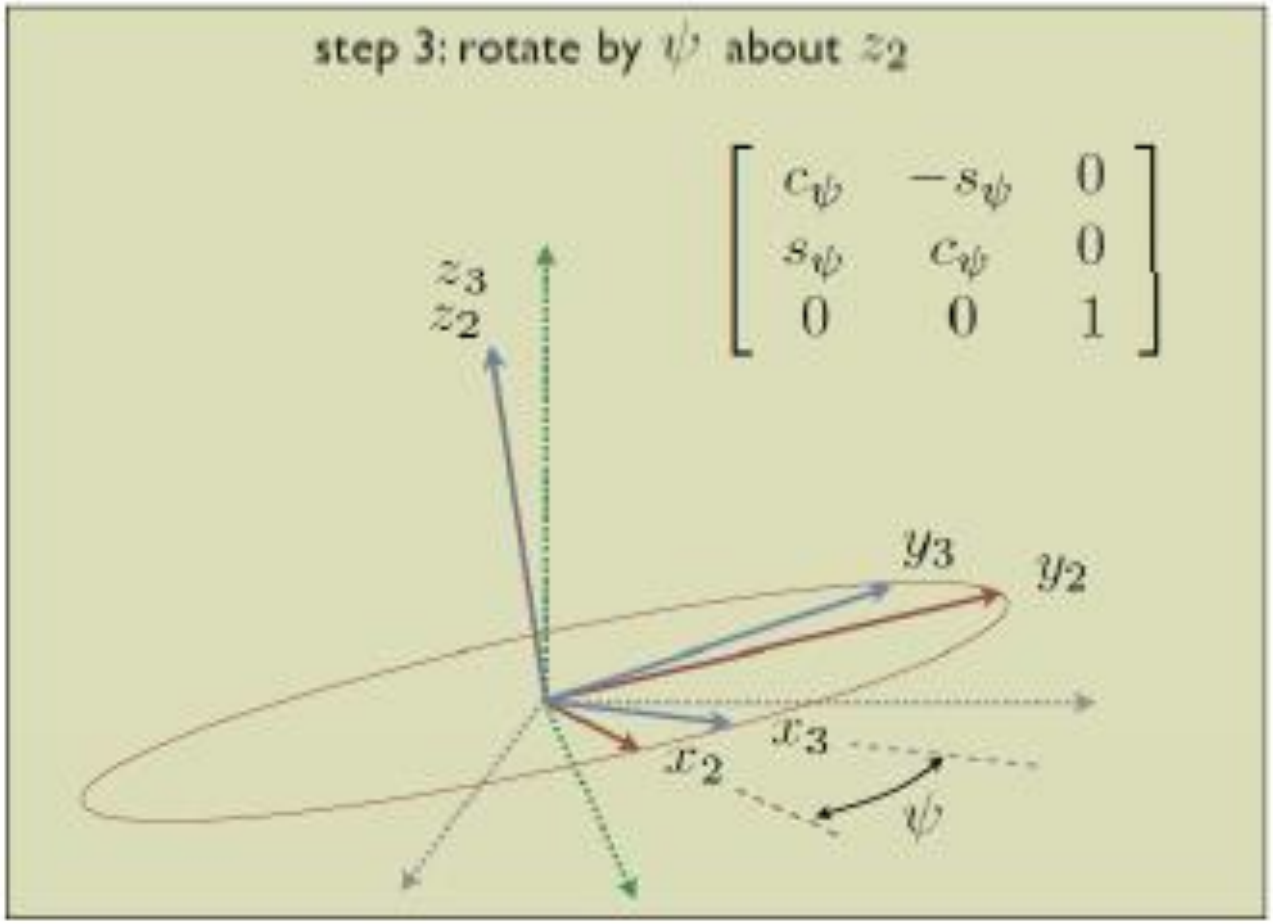
Euler Angles



Euler Angles



Euler Angles



Euler Angles to Rotation Matrix

(**post**-multiply using the **basic rotation matrices**)

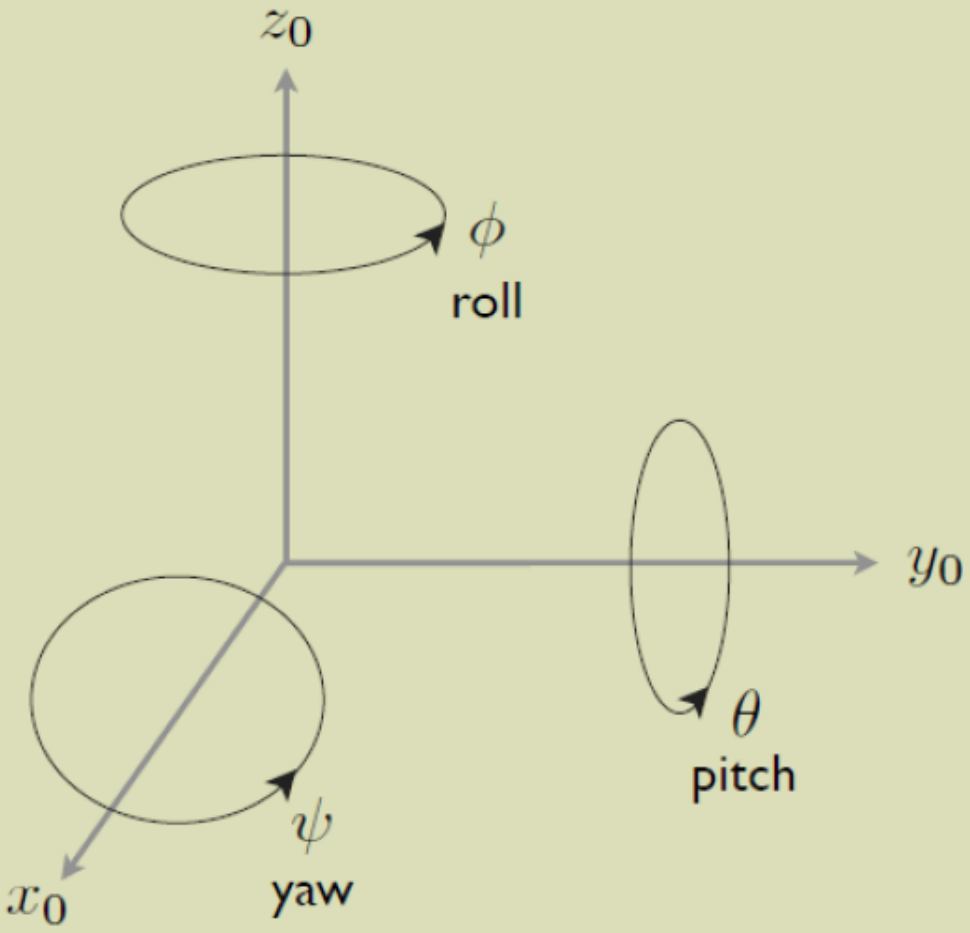
$$\mathbf{R} = \mathbf{R}_{z,\phi}\,\mathbf{R}_{y,\theta}\,\mathbf{R}_{z,\psi}$$

$$= \begin{bmatrix} c_\phi & -s_\phi & 0 \\ s_\phi & c_\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix} \begin{bmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} c_\phi c_\theta c_\psi - s_\phi s_\psi & -c_\phi c_\theta s_\psi - s_\phi c_\psi & c_\phi s_\theta \\ s_\phi c_\theta c_\psi + c_\phi s_\psi & -s_\phi c_\theta s_\psi + c_\phi c_\psi & s_\phi s_\theta \\ -s_\theta c_\psi & s_\theta s_\psi & c_\theta \end{bmatrix}$$

Roll, Pitch, Yaw Angles

defined as a set of three angles about a **fixed** reference



Roll, Pitch, Yaw Angles to Rotation Matrix

(**pre**-multiply using the **basic rotation matrices**)

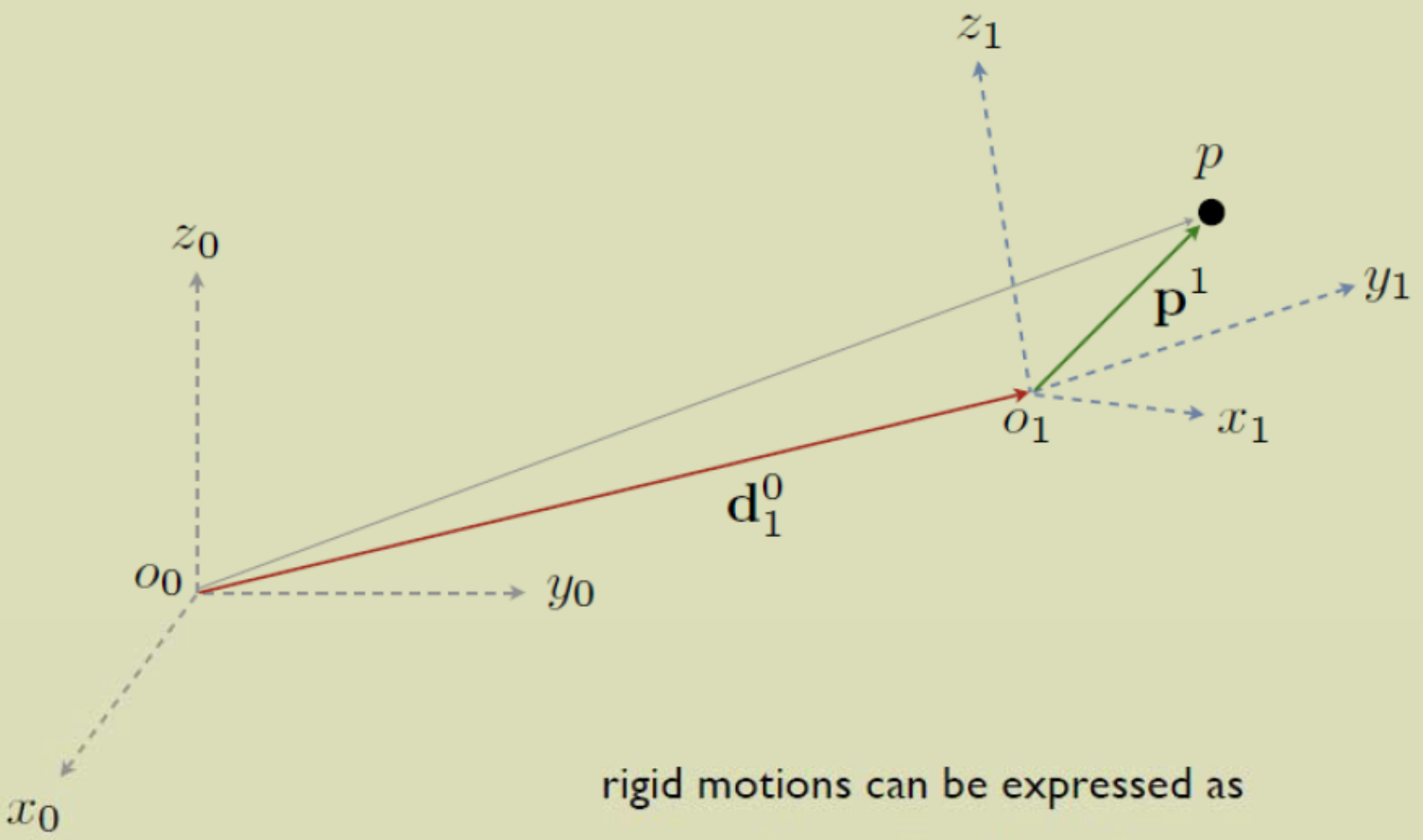
$$\mathbf{R} = \mathbf{R}_{z,\phi}\,\mathbf{R}_{y,\theta}\,\mathbf{R}_{x,\psi}$$

$$= \begin{bmatrix} c_\phi & -s_\phi & 0 \\ s_\phi & c_\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\psi & -s_\psi \\ 0 & s_\psi & c_\psi \end{bmatrix}$$

$$= \begin{bmatrix} c_\phi c_\theta & c_\phi s_\theta s_\psi - s_\phi c_\psi & s_\phi s_\psi + c_\phi s_\theta c_\psi \\ s_\phi c_\theta & s_\phi s_\theta s_\psi + c_\phi c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi \\ -s_\theta & c_\theta s_\psi & c_\theta c_\psi \end{bmatrix}$$

Rigid Motion

a **rigid motion** couples pure translation with pure rotation



rigid motions can be expressed as

$$\mathbf{p}^0 = \mathbf{R}_1^0\,\mathbf{p}^1 + \mathbf{d}_1^0$$

Homogeneous Transformation

a **homogeneous transform** is a matrix representation of rigid motion, defined as

$$\mathbf{H} = \begin{bmatrix} \mathbf{R} & \mathbf{d} \\ \mathbf{0} & 1 \end{bmatrix}$$

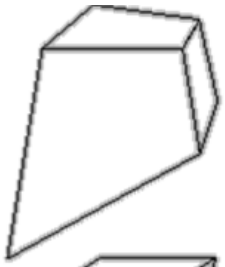
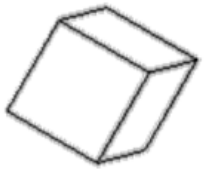
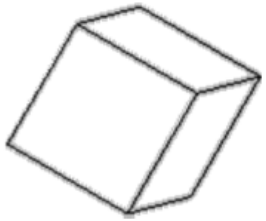
where $\mathbf{R}$ is the 3x3 rotation matrix, and $\mathbf{d}$ is the 1x3 translation vector

$$\mathbf{H} = \begin{bmatrix} n_x & s_x & a_x & d_x \\ n_y & s_y & a_y & d_y \\ n_z & s_z & a_z & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

the **inverse** of a homogeneous transform can be expressed as

$$\mathbf{H}^{-1} = \begin{bmatrix} \mathbf{R}^\top & -\mathbf{R}^\top d \\ 0 & 1 \end{bmatrix}$$

Hierarchy of 3D Transformations

Projective 15dof	$\begin{bmatrix} A & t \\ v^T & v \end{bmatrix}$		Intersection and tangency
Affine 12dof	$\begin{bmatrix} A & t \\ 0^T & 1 \end{bmatrix}$		Parallellism of planes, Volume ratios, centroids, **The plane at infinity π_∞**
Similarity 7dof	$\begin{bmatrix} sR & t \\ 0^T & 1 \end{bmatrix}$		Angles, ratios of length **The absolute conic Ω_∞**
Euclidean 6dof	$\begin{bmatrix} R & t \\ 0^T & 1 \end{bmatrix}$		Volume

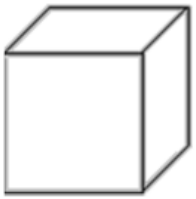
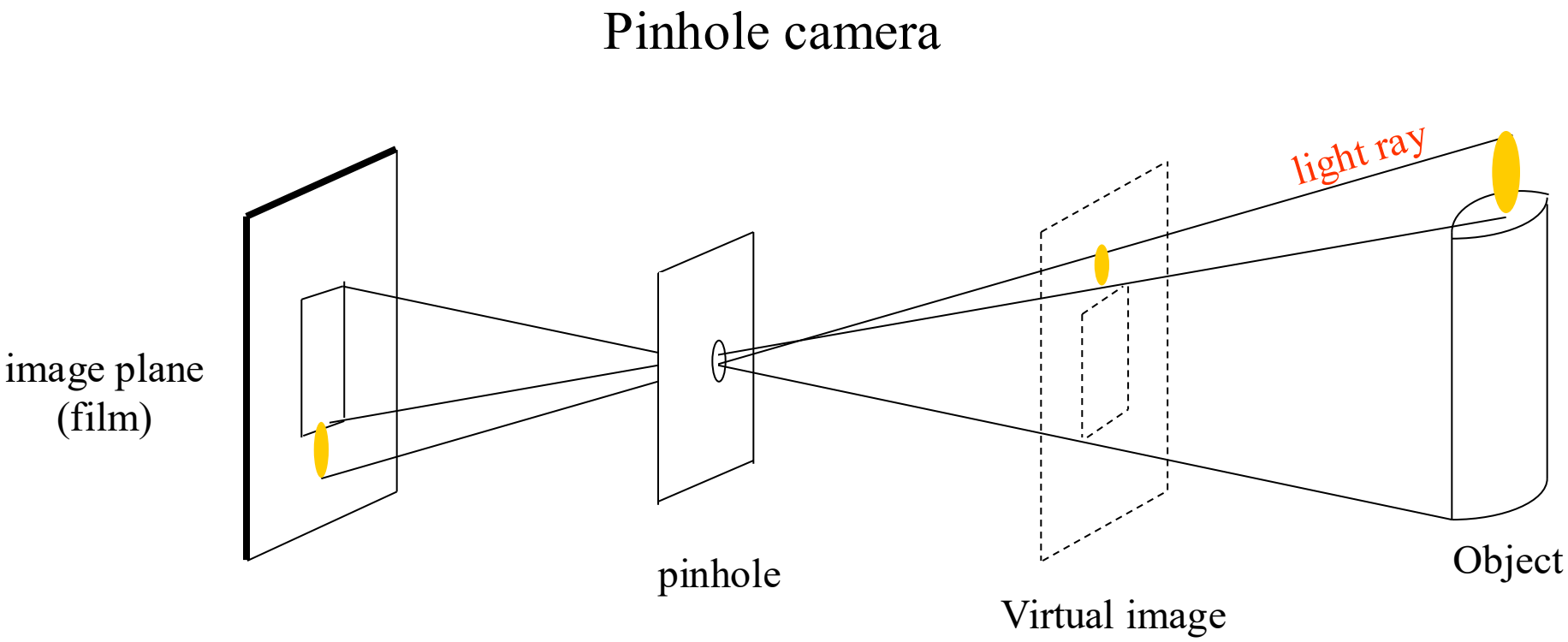


Image Formation

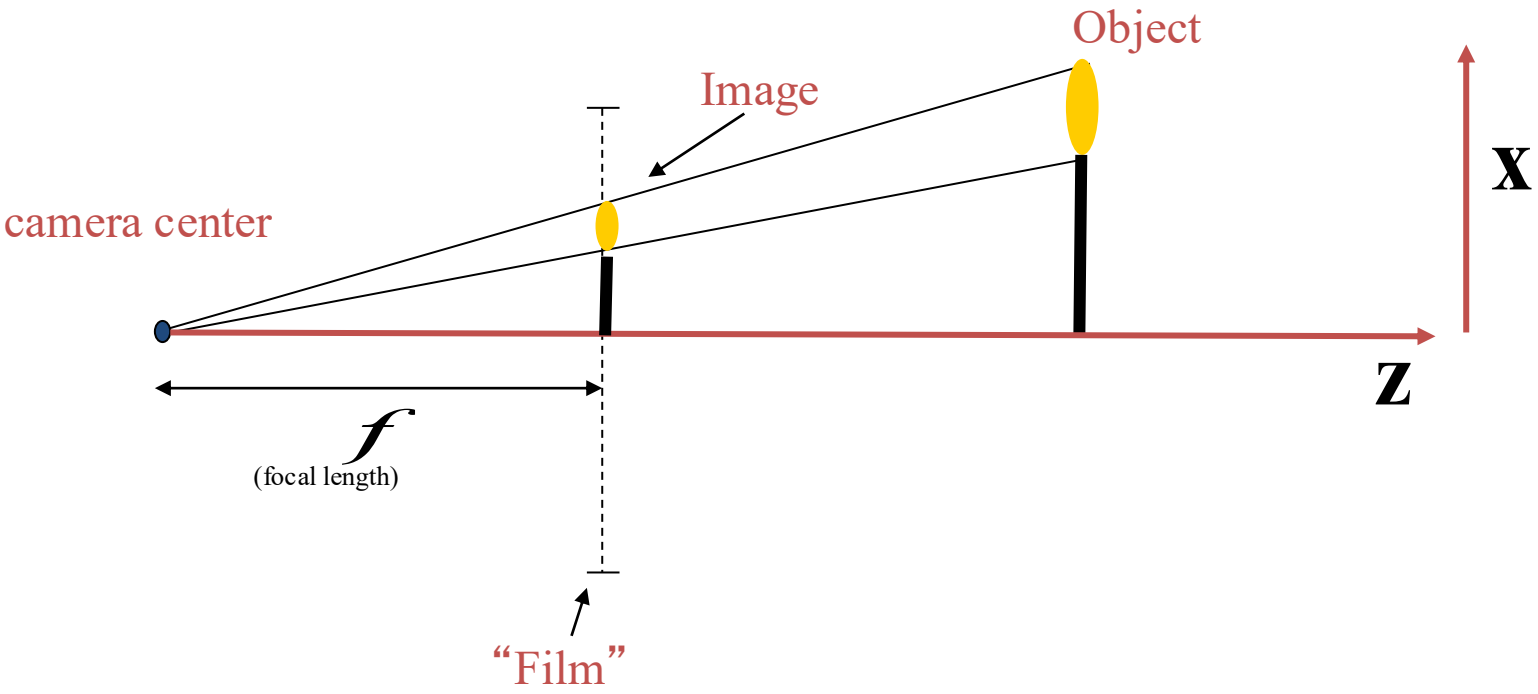
Based on slides by John Oliensis

Image Formation

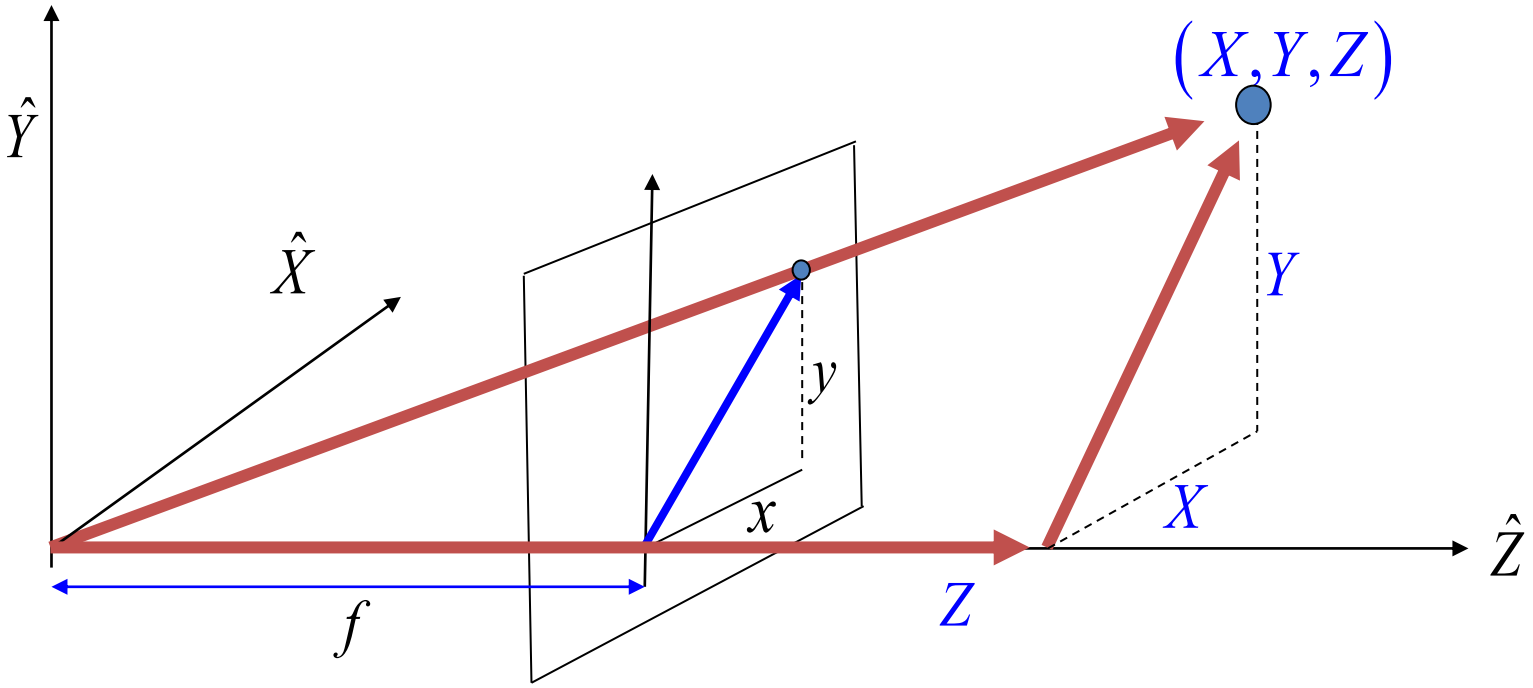


Projection Equation

- 2D world ➔ 1D image



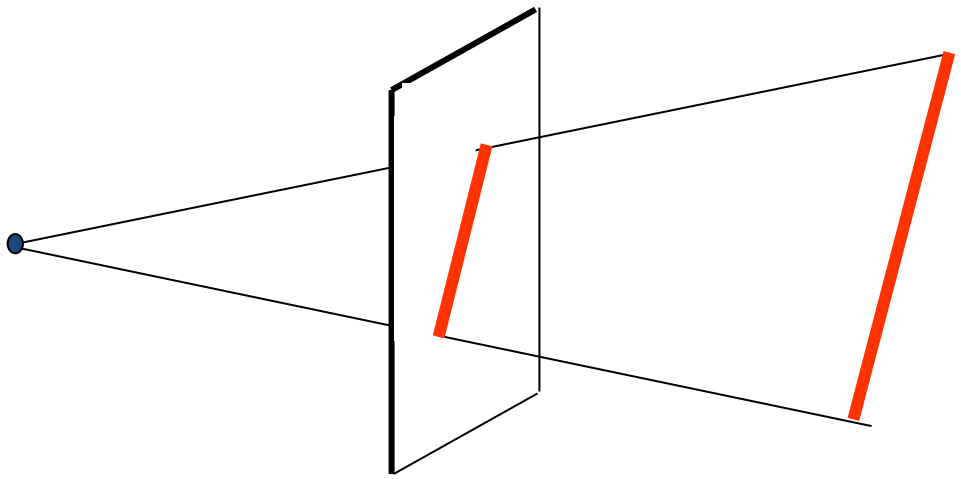
Projection Equation: 3D



Similar triangles: $\frac{x}{X} = \frac{y}{Y} = \frac{f}{Z}$ $\longrightarrow$ $(x, y) = \frac{f}{Z}(X, Y)$

Perspective Projection: Properties

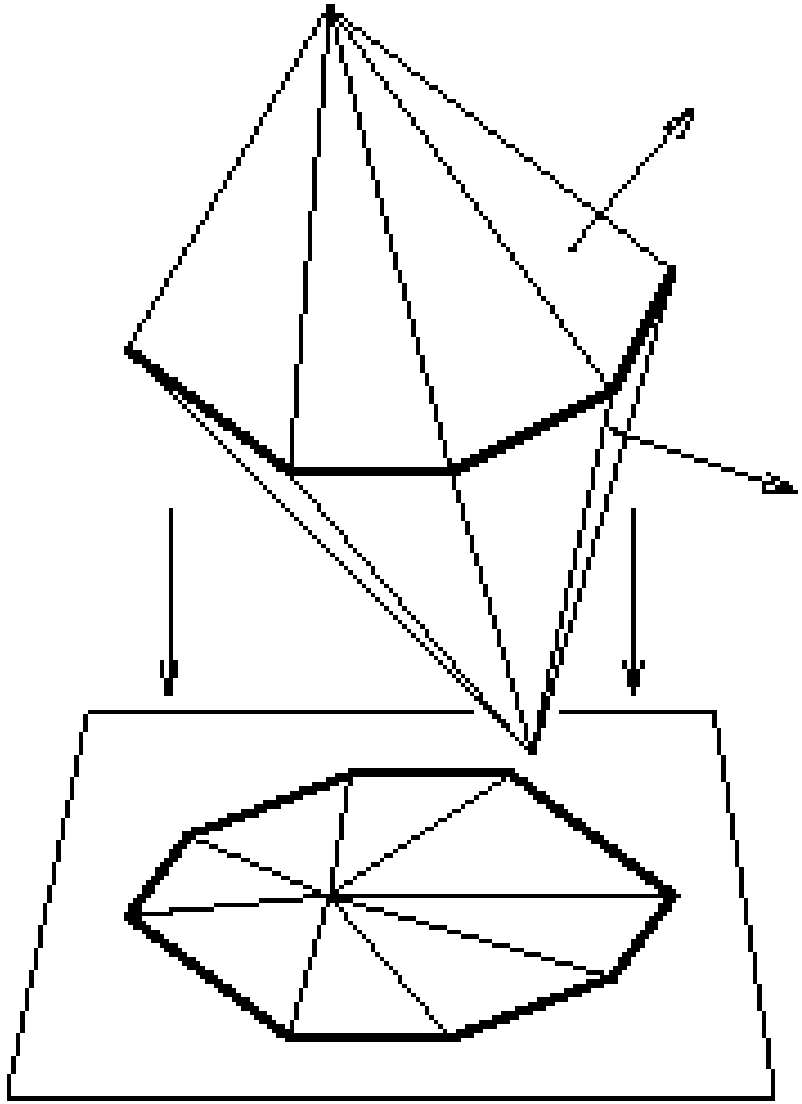
- 3D points ➔ image points
- 3D straight lines ➔ image straight lines



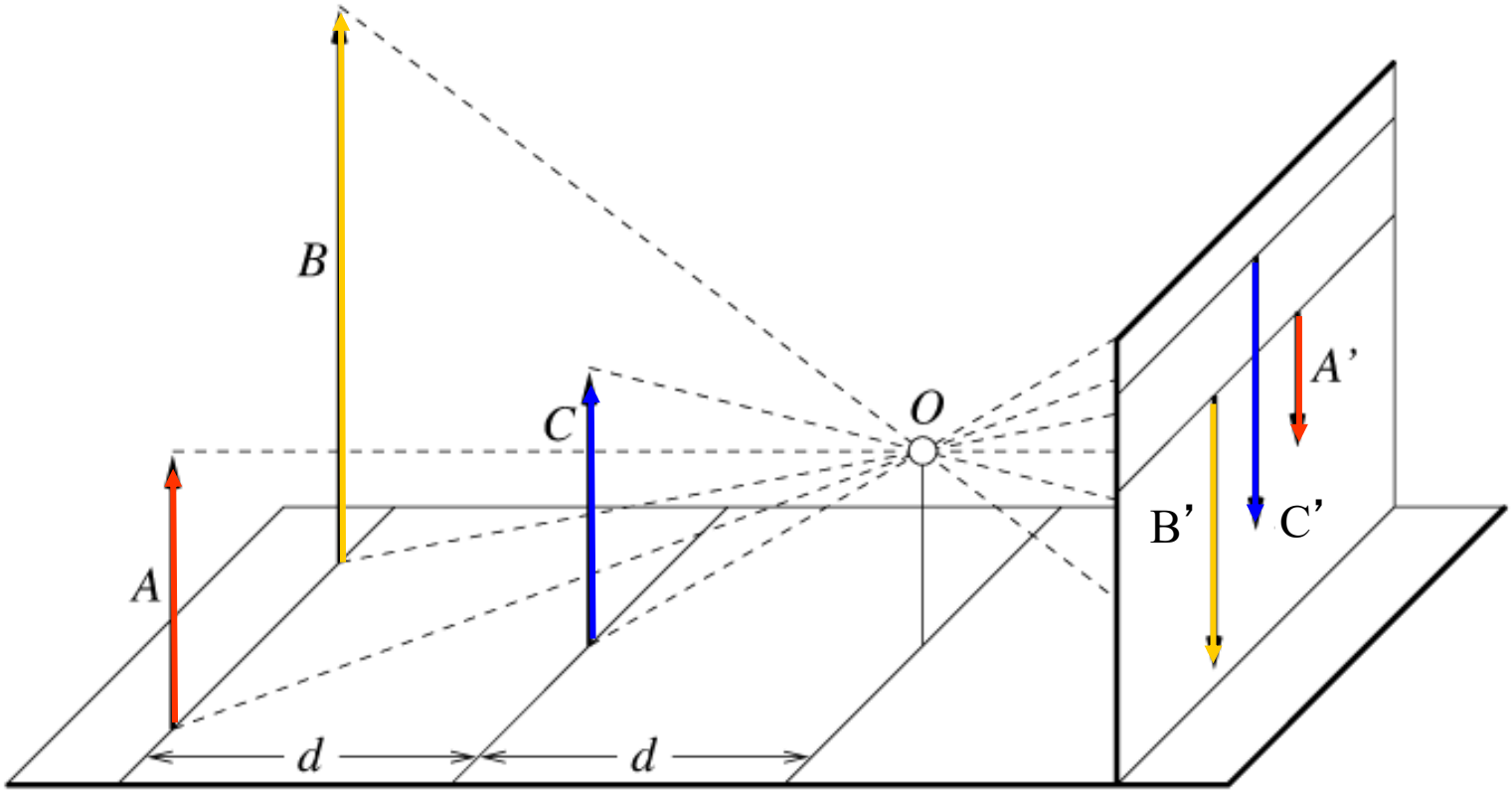
- 3D Polygons ➔ image polygons

Polyhedra Project to Polygons

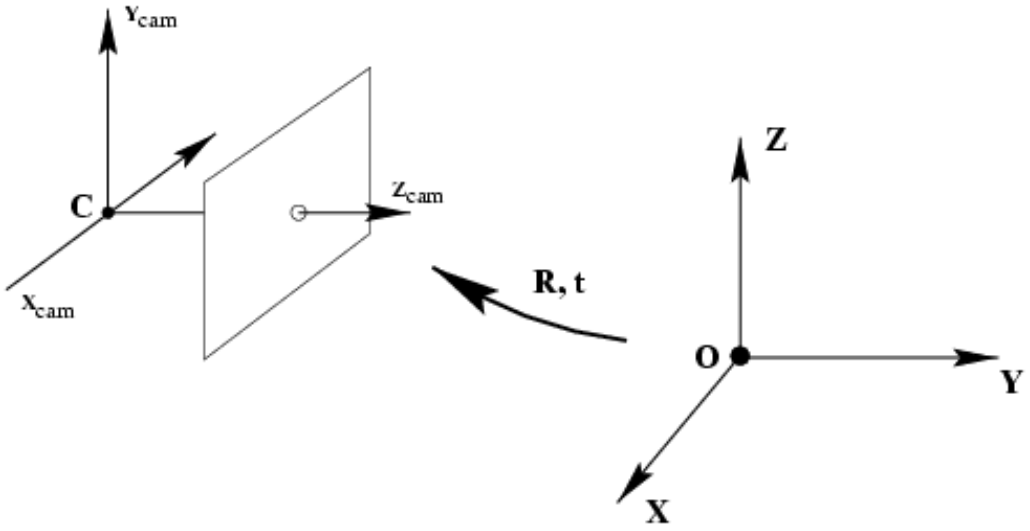
(since lines project to lines)



Properties: Distant objects are smaller



Extrinsic Parameters



Scene motion

$$M = \begin{bmatrix} R_{(3x3)} & t_{(3x1)} \\ 0_{(1x3)} & 1 \end{bmatrix}$$

Camera motion

$$M' = \begin{bmatrix} R^{T}_{(3x3)} & -(R^{T}t)_{3x1} \\ 0_{(1x3)} & 1 \end{bmatrix}$$

Projection matrix

- Includes coordinate transformation and camera intrinsic parameters

$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

- Everything we need to know about a pinhole camera
- Unambiguous
- Can be decomposed into parameters

Projection matrix

- Mapping from 2-D to 3-D is a function of internal and external parameters

$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R^{\mathsf{T}} \mid -R^{\mathsf{T}} t \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\lambda x = K \begin{bmatrix} R^{\mathsf{T}} \mid -R^{\mathsf{T}} t \end{bmatrix} X$$

$$\lambda x = PX$$

Hands On: Camera Motion

- Choose a few 3D points visible to a camera at the origin. (f=500, w=500, h=500)
- Now, move the camera by 2 units of length on the z axis. What happens to the images of the points?
- Rotate the points by 45 degrees about the z axis of the camera and then translate them by 5 units on the z axis away from the camera. What are the new images of the points?